Linear Algebra Done Right

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Exercise Workings

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Contents

I plan to mostly take notes on paper, but I will a lot of the exercises here because I need to practice writing LaTeX quickly and stuff. If you combine both physical notes and these exercises, I produce pages at approximately 50% the rate that I read them, which is a lot! It's split about 20% for exercises and 30% for notes.

As for choosing which exercises to do, as a baseline I try to do odd exercises; if an odd exercise is quite boring I might swap it out for the corresponding even exercise or skip that pair of exercises entirely. All (odd) problems have been thought through, however.

The rigor of my proofs varies a lot, especially if I think a problem is pretty simple, but every subchapter there's always a few problems that are hard enough for me to try with "full" rigor and whatnot.

Even if it is not a proof, all answers will be wrapped in the proof environment (beginning with *Proof.* and ending with □)

1 Vector Spaces

1.A $\mathbb{R}^n$ and $\mathbb{C}^n$

These exercises are really easy but I am always scared that my proofs of these are not rigorous enough.... I do odds.

1 Show that $\alpha + \beta = \beta + \alpha$ for all $\alpha, \beta \in \mathbb{C}$.

Proof. Let $\alpha = a + bi$ and $\beta = c + di$ where $a, b, c, d \in \mathbb{R}$.

Then we have

$$\begin{aligned} \alpha + \beta = \\ a + bi + c + di = \\ (a + c) + (b + d)i = \end{aligned}$$

Because addition is commutative under the reals, we can swap the terms.

$$\begin{aligned} (c + a) + (d + b)i = \\ c + di + a + bi = \\ \beta + \alpha \end{aligned}$$

□

3 Show that $(\alpha\beta)\lambda = \alpha(\beta\lambda)$ for all $\alpha, \beta, \lambda \in \mathbb{C}$.

Distribute, multiply, then rearrange terms using the communitivity of the reals to get the answer. It is quite tedious, so I won't do it here.

5 Show that for every $\alpha \in \mathbb{C}$, there exists a unique $\beta \in \mathbb{C}$ such that $\alpha + \beta = 0$.

Proof. Let $\alpha = a + bi$. Then, let $\beta = -a - bi$. We can observe that

$$\begin{aligned} \alpha + \beta = \\ a + bi - a - bi = \\ (a - a) + i(b - b) = 0 \end{aligned}$$

□

7 Show that $\frac{-1+\sqrt{3}i}{2}$ is a cube root of 1 (meaning that its cube equals 1)

Again, this is a tedious problem where you just multiply it out. Of course, I can represent this number as $e^{\frac{\pi}{3}i}$ but that's cheating.

9 Find $x \in \mathbb{R}^4$ such that

$(4, -3, 1, 7) + 2x = (5, 9, -6, 8)$

Proof. This is vector addition, so we can apply the definition.

$$\begin{aligned}(4,-3,1,7)+2x &= (5,9,-6,8)\\ (4+2x_1,-3+2x_2,1+2x_3,7+2x_4) &= (5,0,-6,8)\end{aligned}$$

For each term in the list, the number must hold, so in essence we get four different equations:

$$\begin{aligned}4+2x_1 &= 5\\ -3+2x_2 &= 0\\ 1+2x_3 &= -6\\ 7+2x_4 &= 8\end{aligned}$$

Each of these linear equations yields only one solution, so the only solution for the equation is $\boxed{x=(\frac{1}{2},-\frac{3}{2},-\frac{7}{2},\frac{1}{2})}$ ∎

1.B Definition of a Vector Space

This one introduces the concept of a vector space in an abstract manner and gives a few unorthodox examples.

1 Prove that $-(-v)=v$ for every $v\in V$

Proof. $-(-v)$ is the additive inverse of $-v$. Therefore,

$$\begin{aligned}-(-v)+(-v) &= 0\\ -(-v)+(-v+v) &= v\\ -(-v)+0 &= v\\ -(-v) &= v\end{aligned}$$

As desired. ∎

3 Suppose $v,w\in V$. Explain why there exists a unique $x\in V$ such that $v+3x=w$.

$$\begin{aligned}v+3x &= w\\ 3x &= w-v\\ x &= \frac{1}{3}(w-v)\end{aligned}$$

Since that right side is a unique vector, x is a unique vector. You invoke the properties and stuff for this, but I'm too lazy.

5 Show that in the definition of a vector space (1.20), the additive inverse condition can be replaced with the condition that

$$0v=0 \text{ for all } v\in V.$$

Here the 0 on the left side is the number 0, and the 0 on the right side is the additive identity of V.

Proof. We can show that given this condition, every element $v \in V$ has an additive inverse.

$$0v = 0$$
$$(1 + -1)v = 0$$

Note that even though we don't know whether *vector* addition has an additive inverse the 0 on the left size is just a scalar, so we can still say there is an additive inverse.

Then distribute and apply the multiplicative identity axiom...

$$1v + (-1)v = 0$$
$$v + (-1)v = 0$$

Therefore $(-1)v$ is the additive inverse for all $v \in V$. □

7 Suppose S is a nonempty set. Let V^S denote the set of functions from S to V. Define a natural addition and scalar multiplication on V^S , and show that V^S is a vector space with these definitions.

Proof. Like with F^S, we can sort of "piggyback" off the properties of V to maintain the properties of a vector space in V^S.

Define addition as

$$(f + g)(x) = f(x) + g(x)$$

Let the additive identity be $0(x) = 0$ for all $x \in S$.

$$(f + 0)(x) = f(x) + 0(x) = f(x) + 0 = f(x)$$

For all $f, g \in V^S$. Because $f(x)$ is a vector, it has an inverse (namely $-f(x)$). Therefore, we can define the additive inverse as

$$(-f)(x) = -f(x)$$

$$(f + (-f))(x) = f(x) + (-f(x)) = 0 = 0(x)$$

Communativity:

$$(f + g)(x) = f(x) + g(x) = g(x) + f(x) = (g + f)(x)$$

We can swap $f(x)$ and $g(x)$ because V is communative.

Associativity:

$$((f + g) + h)(x) = (f(x) + g(x)) + h(x) = f(x) + (g(x) + h(x)) = (f + (g + h))(x)$$

Again, we use the properties of the vector space V to help us out here.

Scalar multiplication

For all $\lambda \in \mathbb{F}, v \in V^S$,

$$(\lambda v)(x) = \lambda v(x)$$

Multiplicative Identity

1 is the multiplicative identity.

$$(1v)(x) = 1v(x) = v(x)$$

Since 1 is the multiplicative identity of V.

Distributive property

$$(a(u+v))(x) = a(u+v)(x) = a(u(x)+v(x)) = au(x)+av(x) = (au+av)(x)$$

Because of the distributive property of V.

$$((a+b)v)(x) = (a+b)v(x) = av(x)+bv(x) = (av+bv)(x)$$

For the same reason.

Closure

Because V is closed under addition and scalar multiplication, and all our operations are defined using these 2 operations, V^S is closed.

Thus, we have proved all the properties of vector spaces on V^S, therefore V^S is a vector space.

□

1.C Subspaces

This is where it ramps up a little bit, we are doing some linear algebra now! This idea of "subspaces" is interesting...

1 For each of the following subsets of $\mathbb{F}^3$, determine whether it is a subspace of $\mathbb{F}^3$

(a) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 0\}$

Proof. $(0, 0, 0)$ is in our set, so we have the additive identity. Consider vectors x and y in our set, and add them.

$$\begin{aligned} x + y &= \\ (x_1, x_2, x_3) + (y_1, y_2, y_3) &= \\ (x_1 + y_1, x_2 + y_2, x_3 + y_3) &= \\ z &= \\ z_1 + 2z_2 + 3z_3 &= \\ x_1 + y_1 + 2x_2 + 2y_2 + 3x_3 + 3y_3 &= \\ x_1 + 2x_2 + 3x_3 + y_1 + 2y_2 + 3y_3 &= \\ 0 + 0 &= 0 \end{aligned}$$

Because x and y are members of the subset, we can use the condition. Therefore, z is part of our set.

Now let us consider scalar multiplication. Consider x in our set and $\lambda \in \mathbb{F}$. Then,

$$\begin{aligned} \lambda x &= \\ (\lambda x_1, \lambda x_2, \lambda x_3) \end{aligned}$$

We then consider whether this new element is in our set.

$$\begin{aligned} \lambda x_1 + 2\lambda x_2 + 3\lambda x_3 = \\ \lambda(x_1 + 2x_2 + 3x_3) = \\ \lambda(0) = 0 \end{aligned}$$

We use the distributive property here. Since it follows the set's condition, we can say that it is closed under scalar multiplication.

These are all the conditions, so we can say that our set is a subspace of $\mathbb{F}^3$. $\square$

(b) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 4\}$

Proof. Consider the element $(1, 0, 1)$ which is in our set because $1 + 2 \cdot 0 + 3 \cdot 1 = 4$ and it is in $\mathbb{F}^3$. Now consider multiplying it by the scalar 2 to create the element $(2, 0, 2)$. Unfortunately, $2 + 2 \cdot 0 + 3 \cdot 2 = 8 \neq 4$, so this set does not fulfill the closure under scalar multiplication condition. $\square$

(c) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 x_2 x_3 = 0\}$

Proof. Consider the sets $(1, 1, 0)$ and $(0, 0, 1)$. Summing these up, we get $(1, 1, 1)$, but $1 \cdot 1 \cdot 1 = 1 \neq 0$, so this set fails the additive closure property. $\square$

(d) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 = 5x_3\}$

I am too lazy but I did the other 3, good enough

2 I did number 2 in my notebook, not noticing it was actually an exercise.

3 Show that the set of differentiable real-valued functions f on the interval $(-4, 4)$ such that $f'(-1) = 3f(2)$ is a subspace of $\mathbf{R}^{(-4,4)}$.

Proof. Clearly this set is a subset of $\mathbb{R}^{(-4,4)}$. The zero function ($f(x) = 0$) satisfies the condition (since the derivative is also 0), so we have the additive identity.

Consider functions f, g in the set, then consider $(f + g)(x)$. Functions and their derivatives are additive, so we can write

$$\begin{aligned} (f+g)'(-1) = \\ f'(-1) + g'(-1) = \\ 3f(2) + 3g(2) = \\ 3(f(2) + g(2)) = \\ 3(f+g)(2) \end{aligned}$$

Therefore, we have additive closure.

Consider f in the set and $\lambda \in \mathbb{R}$. Let us show that $(\lambda f)(x) = \lambda f(x)$ is in the set.

$$\begin{aligned} (\lambda f)'(-1) = \\ \lambda f'(-1) = \\ 3\lambda f(2) = \\ 3(\lambda f)(2) \end{aligned}$$

So we are closed under scalar multiplication. $\square$

5 Is $\mathbb{R}^2$ a subspace of the complex vector space $\mathbb{C}^2$?

Proof. Although $\mathbb{R}^2$ is a vector space over $\mathbb{R}$, it is not a vector space over $\mathbb{C}$. For example, multiplying $(1,1)$ by i produces a vector not in $\mathbb{R}^2$, so it is not closed under scalar multiplication. $\square$

7 If U is closed under addition and under taking additive inverses (meaning $-u \in U$ whenever $u \in U$), then U is a subspace of $\mathbb{R}^2$.

Proof. Consider the set $\mathbb{Q}^2$. It is closed under addition and additive inverse, but it is *not* closed under scalar multiplication over $\mathbb{R}$. For example, the set $(1,1)$ when multiplied by an irrational number such as $\sqrt{2}$ is not a part of the set. $\square$

9 A function $f : \mathbb{R} \to \mathbb{R}$ is called periodic if there exists a positive number p such that $f(x) = f(x+p)$ for all $x \in \mathbb{R}$. Is the set of periodic functions from $\mathbb{R}$ to $\mathbb{R}$ a subspace of $\mathbb{R}^{\mathbb{R}}$? Explain.

Proof. There is not closure under addition, unfortunately. Consider the functions

$$f(x) = \cos(2\pi x)$$
$$g(x) = \cos(2x)$$

$f(x)$ is at its maximum of 1 whenever $x \in \mathbb{Z}$, and $g(x)$ is at its maximum of 1 whenever $x = n\pi, n \in \mathbb{Z}$. $(f+g)(x) = 2$ only when both of these conditions are met. However, this is only true when $x = 0$.

Suppose that $x = n\pi$ and $x = m$ where $n, m \in \mathbb{Z}$. Then,

$$n\pi = m$$
$$\pi = \frac{m}{n}$$

Which implies that π is rational if $n \neq 0$, so the only solution is $x = 0$.

Therefore, $(f+g)(x)$ reaches its maximum only at $x = 0$, and there is no $p > 0$ where $(f+g)(0) = (f+g)(p)$. Therefore $(f+g)(x)$ is outside the set and it fails the additive closure condition, making it not a vector space. $\square$

11 Prove that the intersection of every collection of subspaces of V is a subspace of V.

Proof. Consider subspaces of V, $V_1, \ldots, V_M$. All subspaces contain the identity element, so $V_1 \cap \ldots \cap V_M$ will also contain it. Consider $u, v \in V_1 \cap \ldots V_M$. Because $u, v \in V_k$, any addition and scalar multiplication will remain in V_k. Therefore, any addition and scalar multiplication will remain in $V_1 \cap \ldots \cap V_M$, completing the proof. $\square$

12 Prove that the union of two subspaces of V is a subspace of V if and only if one of the subspaces is contained in the other.

First, lets prove a lemma we will need for this.

Lemma. *Given $u \notin V, v \in V, u + v \notin V$.*

Proof. Suppose that $u + v \in V$. Then, because of the additive inverse, $u + v - v = u \in V$, and we reach a contradiction. $\square$

Now we can prove the main result.

Proof. Consider subspaces V_1, V_2

Consider WLOG $V_1 \subseteq V_2$. Then, $V_1 \cup V_2 = V_2$, and since V_2 is a subspace so is $V_1 \cup V_2$.

The converse is trickier. We prove the contrapositive. Consider when V_1 and V_2 are not subsets of each other. Then, there is some element $u \in V_1, u \notin V_2$. There must also be an element $v \in V_2, v \notin V_1$. These are true from the definition of subset. Consider $u + v$. By the lemma, this sum is in neither set, so it is not in the union of V_1 and V_2, so $V_1 \cup V_2$ is not a subspace.

□

13 Prove that the union of three subspaces of V is a subspace of V if and only if one of the subspaces contains the other two.

This problem is very difficult (as the textbook alluded) and I needed to look up the solution.

Proof. If any subspace is contained within another this reduces to the 2-subspace case.

Presume that $V_1 \cup V_2 \cup V_3$ is a subspace.

Suppose $V_1 \cup V_2 \backslash V_3$ was not empty. Then, we can use $u \in V_1 \cup V_2 \backslash V_3$ and $v \in V_3 \backslash V_2 \cup V_1$ to prove that there is not closure using the lemma from exercise 12. By symmetry, there is no element that is in exactly two of the subsets.

From here on, suppose $u \in V_1 \backslash V_2 \cup V_3$, $v \in V_2 \backslash V_1 \cup V_3$. Note that if these elements did not exist one set would be a subset of another or we would have an element in exactly two subsets. Then, to be closed, $u + v \in V_3$ since by the Exercise 12 lemma we cannot have $u + v$ be in either V_1 or V_2. Consider a, b in $\mathbb{F}$ where $a, b \neq 0, a - b \neq 0$. We also have $au + v \in V_3$ and $bu + v \in V_3$ since scalar multiplication by a non-zero constant should not change which subspaces a vector is in, so $au + v - (bu - v) = (a - b)u \in V_3$. However, u was explicitly only in V_1, so we have a contradiction and $V_1 \cup V_2 \cup V_3$ is not actually a subspace! Note that this proof does not work in $\mathbb{F}_2$ because we do not have 2 different numbers that satisfy the properties of a and b. □

15 Suppose U is a subspace of V. What is $U + U$?

Proof. We use the definition of subspace addition.

$$U + U = \{u + v : u, v \in U\}$$

Because U is a subspace and is therefore closed, this is identical to U. □

17 Is the operation of addition on the subspaces of V associative? In other words, if V_1, V_2, V_3 are subspaces of V, is

$$(V_1 + V_2) + V_3 = V_1 + (V_2 + V_3)?$$

Proof. Using the definition of subspace, we have

$$\begin{aligned}(V_1 + V_2) + V_3 &= \{(v_1 + v_2) + v_3 : v_1 \in V_1, v_2 \in V_2, v_3 \in V_3\} \\ &= \{v_1 + (v_2 + v_3)v_1 \in V_1, v_2 \in V_2, v_3 \in V_3\} \\ &= V_1 + (V_2 + V_3)\end{aligned}$$

using the associative property of vector addition. □

19 Prove or give a counterexample: If V_1, V_2, U are subspaces of V such that

$$V_1 + U = V_2 + U$$

then $V_1 = V_2$

Proof. This is a nice application of the idea that subspaces do not have inverses. For example, we could have V as $\mathbb{R}^2$, $V_1 = \{(x, 0) : x \in \mathbb{R}\}$, and $V_1 = \{(0, y) : y \in \mathbb{R}\}$, and $U = V$. These both add to V, but they are not equal. □

21 Suppose

$$U = \{(x, y, x + y, x - y, 2x) \in \mathbf{F}^5 : x, y \in F\}$$

Find a subspace W of $\mathbf{F}^5$ such that $\mathbf{F}^5 = U \oplus W$.

Proof. We have 2 variables, so we should have 3 variables in the other one to counteract this. The first 2 terms of U are just x and y, which is great because now they cannot vary. We can choose

$$W = \{(0, 0, p, q, r) \in \mathbf{F}^5 : p, q, r \in \mathbf{F}\}$$

And then any vector of the form (a, b, c, d, e) using

$$\begin{aligned}
x &= a \\
y &= b \\
p &= c - (x + y) = c - a - b \\
q &= d - (x - y) = c - a + b \\
r &= e - (2x) = e - 2a
\end{aligned}$$

□

23 Prove or give a counterexample: If V_1, V_2, U are subspaces of V such that

$$V = V_1 \oplus U \text{ and } V = V_2 \oplus U$$

then $V_1 = V_2$.

Proof. This is untrue. Choose $V = \mathbb{R}^2$. Let $U = \{(x, 0) : x \in \mathbb{R}\}$. Then choose $V_1 = \{(0, b) : b \in \mathbb{R}\}$ but $V_2 = \{(a, a) : a \in \mathbb{R}\}$.

These are different subspaces, but it can be shown that their direct sum with U is both V. □

24 A function $f : \mathbb{R} \to \mathbb{R}$ is called *even* if

$$f(-x) = f(x)$$

for all $x \in \mathbb{R}$. A function $f : \mathbb{R} \to \mathbb{R}$ is called *odd* if

$$f(-x) = -f(x)$$

for all $x \in \mathbb{R}$. Let V_e denote the set of real-valued even functions on $\mathbb{R}$ and let V_o denote the set of real-valued odd functions on $\mathbb{R}$. Show that

$$\mathbb{R}^{\mathbb{R}} = V_e \oplus V_o$$

Proof. Given some function $f : \mathbb{R} \to \mathbb{R}$ we can construct it by adding the following functions:

We have $g \in V_e$,

$$g(x) = \frac{f(x) + f(-x)}{2}$$

It is clearly even by symmetry.

We can then choose $h \in V_o$,

$$h(x) = \frac{f(x) - f(-x)}{2}$$

This is clearly odd.

Now we must show that these are the *only* two functions that create f. Given some g and h, we must have $f(x) - g(x) = h(x)$. This means that

$$\begin{aligned}
f(-x) - g(-x) &= -h(x) \\
f(-x) - g(x) &= -h(x) \\
f(-x) &= g(x) - h(x) \\
f(x) &= g(x) + h(x) \\
f(-x) + f(x) &= 2g(x) \\
g(x) &= \frac{f(-x) + f(x)}{2}
\end{aligned}$$

h comes naturally afterward.

$\square$

2 Finite Dimensional Vector Spaces

We say goodbye to our infinite dimensional vector spaces (noooo not $\mathbb{R}^{\mathbb{R}}$) and we now focus on vector spaces which can actually be spanned by a list of vectors. This unit is... harder than the first one definitely, and it feels more like "linear algebra" than just "abstract algebra."

2.A Span and Linear Dependence

Not too bad.

1 Find a list of four distinct vectors in $\mathbf{F}^3$ whose span equals

$$\{(x, y, z) \in \mathbf{F}^3 : x + y + z = 0\}$$

Proof. Actually, I only need 2 vectors for this.

$$v_1 = (1, -1, 0) \text{ and } v_2 = (0, 1, -1)$$

With these, I can construct any vector that satisfies the properties. I can include some other ones but I am lazy.

□

3 Suppose $v_1, \ldots, v_m$ is a list of vectors in V. For $k \in \{1, \ldots, m\}$, let

$$w_k = v_1 + \cdots + v_k$$

Show that $\text{span}(v_1, \ldots, v_m) = \text{span}(w_1, \ldots, w_m)$.

Proof. Given an arbitrary linear combination of w_k,

$$\sum_{k=1}^{m} a_k w_k = \sum_{k=1}^{m} (\sum_{i=1}^{k} a_i) v_k$$

For a given v_k, we can construct it from a linear combination of w as $v_k = w_k - w_{k-1}$ where $v_1 = w_1$. Thus, since we can go both ways, the spans are equal.

□

5 Find a number t such that

$$(3, 1, 4), (2, -3, 5), (5, 9, t)$$

is not linearly independent in $\mathbf{R}^3$

Proof. If we can find some

$$a(3, 1, 4) + b(2, -3, 5) = (5, 9, t)$$

then our list of vectors in not linearly independent. This turns into these equations:

$$\begin{aligned} 3a + 2b &= 5 \\ a - 3b &= 9 \\ 4a + 5b &= t \end{aligned}$$

We solve, and we get that $a = 3, b = -2, t = 2$, so it is not linearly independent when $\boxed{t = 2}$. □

7 (a) Show that if we think of $\mathbb{C}$ as a vector space over $\mathbb{R}$, then the list $1 + i, 1 - i$ is linearly independent.

Proof. Clearly there is no positive multiple of $1 + i$ that creates $1 - i$ you dumbass. The real and imaginary parts will always have the same sign. □

(b) Show that if we think of $\mathbb{C}$ as a vector space over $\mathbb{C}$, then the list $1+i, 1-i$ is linearly dependent.

Proof. We can multiple $1 + i$ by

$$\begin{aligned} \frac{1-i}{1+i} &= \frac{(1-i)(1-i)}{(1+i)(1-i)} \\ &= \frac{-2i}{2} \\ &= -i \end{aligned}$$

to get $1 - i$, therefore its linearly dependent. □

9 Prove or give a counterexample: If $v_1, v_2, \ldots, v_m$ is a linearly independent list of vectors in V, then

$$5v_1 - 4v_2, v_2, v_3, \ldots, v_m$$

is linearly independent.

Proof. Suppose that $5v_1 - 4v_2, v_2, v_3, \ldots, v_m$ was linearly dependent. Then,

$$a_1(5v_1 - 4v_2) + a_2v_2 + \cdots a_mv_m = 5a_1v_1 + (a_2 - 4a_1)v_2 + \cdots + a_mv_m = 0$$

and we have a contradiction. □

11 Prove or give a counterexample: If $v_1, \ldots, v_m$ and $w_1, \ldots w_m$ are linearly independent lists of vectors in V, then the list $v_1 + w_1, \ldots, v_m + w_m$ is linearly independent.

Proof. We can construct a counterexample. Let us use $\mathbb{R}^1$ here. 3 is linearly independent, and so is -3, but $3 - 3 = 0$, which is not linearly independent. □

13 Suppose $v_1, \dots v_m$ is linearly independent in V and $w \in V$. Show that

$v_1, \dots v_m$, w is linearly independent $\iff$ $w \notin \operatorname{span}(v_1, \dots, v_m)$.

Proof. We prove the contrapositive in both directions.

Suppose that $w \in \operatorname{span}(v_1, \dots, v_m)$. Then, there is some linear combination of $v_1, \dots, v_m$ equal to w and so $v_1, \dots, v_m, w$ is linearly dependent.

Suppose that $v_1, \dots, v_m, w$ is linearly dependent. Then, by the linear dependence lemma, there exists some element in this list that is in the span of the previous ones. Because all lists $v_1, \dots v_k, k \leq m$ are linearly independent, $w \in \operatorname{span}(v_1, \dots, v_m)$. □

15 Explain why there does not exist a list of six polynomials that is linearly independent in $\mathcal{P}_4(\mathbf{F})$.

Proof. We have a list of 5 polynomials that spans $\mathcal{P}_4(\mathbf{F})$. The length of a linearly independent list must be less than or equal to the length of this spanning list, so there does not exist a list of six polynomials that is linearly independent in $\mathcal{P}_4(\mathbf{F})$. □

17 Prove that V is infinite-dimensional if and only if there is a sequence $v_1, v_2, \dots$ of vectors in V such that $v_1, \dots, v_m$ is linearly independent for every positive integer m.

Proof. Because the length of the span of a vector space must be greater or equal to any given list of linearly independent vectors (Replacement theorem), this sequence means that there is no finite list of vectors that spans V.

If V is infinite-dimensional, we can construct such a sequence $v_1, v_2, \dots$. Begin by choosing an arbitrary vector $v_1 \neq 0$. Then, for each $k \geq 2$, we know that

$$\operatorname{span}(v_1, \dots, v_{k-1}) \neq V$$

So there exists some $v_k \notin \operatorname{span}(v_1, \dots, v_{k-1}$. Continue for an arbitrary number of steps. □

19 Prove that the real vector space of all continuous real-valued functions on the interval $[0, 1]$ is infinite-dimensional.

Proof. I can construct a sequence of functions that are all linearly independent. In fact, I can use

$$1, x, x^2, x^3 \dots$$

Since each polynomial is uniquely determined by its coefficients, these are linearly independent, and they are also clearly continuous. □

20 Suppose $p_0, p_1, \dots, p_m$ are polynomials in $\mathcal{P}_m(\mathbf{F})$ such that $p_k(2) = 0$ for each $k \in \{0, \dots, m\}$. Prove that $p_0, p_1, \dots, p_m$ is not linearly independent in $\mathcal{P}_m(\mathbf{F})$.

Proof. For $m = 0$, $p_0 = 0$, which isn't linearly independent.

For $m > 0$, if $p_k = 0$ for some $k \in \{0, \dots, m\}$, then our set is already not linearly independent.

Suppose $p_k \neq 0$ for all $k \in \{0, \dots, m\}$. Then, because $p_k(2) = 0$, all p_k has a factor of $(z - 2)$. When finding a linear combination that equals zero, we can factor out all by $(z - 2)$. That is

$$a_1p_1 + a_2p_2 + \cdots + a_mp_m = 0 \iff (z-2)(a_1\frac{p_1}{z-2} + \cdots + a_m\frac{p_m}{z-2}) = 0$$

Now, we have a set of polynomials of degree $m-1$, but we have $m+1$ terms. For $\mathcal{F}_{m-1}(\mathbf{F})$ there is a list of m vectors that span it. Thus, the length of a linearly independent list of vectors must be less than or equal to m. We have $m+1$ terms in our list, so we can determine that our list is linearly dependent. □

2.B Bases

This is more or less a merging of the concepts of span and linear independence and showing their relationships. Pretty easy still.

1. Find all vector spaces that have exactly one basis.

Proof. We have the vector space of size 1 and dimension 0, $\{0\}$. Its only basis is the empty set. In one dimension, we have the basis for $\mathbb{F}_2^1$, the vector space consisting of only one dimension of the field of size 2, which has the basis 1. If the field had more elements than 2, then you would be able to obtain another basis by multiplying one of the elements by some non-zero scalar. If the vector space had more than 1 dimension, we could choose other bases, such as $\{1,1\},\{0,1\}$ for $\mathbb{F}_2^2$. □

3 (a) Let U be the subspace of $\mathbb{R}^5$ defined by

$$U = \{(x_1, x_2, x_3, x_4, x_5) \in \mathbb{R}^5 : \ x_1 = 3x_2, \ x_3 = 7x_4\}.$$

Find a basis of U.

Proof. x_1 and x_4 are useless, so we can just use the standard basis vectors for $\mathbb{R}^3$ except instead using filled in values for x_1 and x_4. For example,

$$\{3,1,0,0,0\}$$
$$\{0,0,7,1,0\}$$
$$\{0,0,0,0,1\}$$

□

5 Suppose V is finite-dimensional and U, W are subspaces of V such that $V = U + W$. Prove that there exists a basis of V consisting of vectors in $U \cup W$.

Proof. If there is a spanning list of vectors in $U \cup W$, then there is also a basis in $U \cup W$ (since every spanning list can be reduced to a basis).

We can thus a basis of U attached to a basis in W as our spanning list. This list spans V because $V = U + W$ implies for every $v \in V, \exists u \in U, w \in W \ s.t. \ v = u + w$, and since we can create any $u + w$ in our new basis, we are done. □

7 Suppose v_1, v_2, v_3, v_4 is a basis of V. Prove that $v_1 + v_2$, $v_2 + v_3$, $v_3 + v_4$, v_4 is also a basis of V.

Proof. Clearly, our given vectors are in the span of $v_1, \ldots, v_4$.

We will show that you can represent $v_1, \ldots v_4$ as linear combinations of our given vectors (which will thus show that they have the same span). After we derive one vector, we can use it in the formula of other vectors (implicitly applying a substitution to keep it in our vector set)

$$\begin{aligned} v_4 &= v_4 \\ v_3 &= (v_3 + v_4) - v_4 \\ v_2 &= (v_2 + v_3) - v_3 \\ v_1 &= (v_1 + v_2) - v_2 \end{aligned}$$

These vectors must also be linearly independent. Suppose that they were not linearly independent. Then, we could remove a vector without changing its span. Now, we have a span of vectors that is *smaller* than a linearly independent list of vectors. Contradiction! □

8 Prove or give a counterexample: If v_1, v_2, v_3, v_4 is a basis of V and U is a subspace of V such that $v_1, v_2 \in U$ and $v_3 \notin U$ and $v_4 \notin U$, then v_1, v_2 is a basis of U.

Proof. Nope! Suppose we are working in $\mathbb{R}^4$ and our basis is

$$(1,0,0,0), (0,1,0,0), (0,0,1,1), (0,0,0,1)$$

We could have a subspace U that admits basis

$$(1,0,0,0), (0,1,0,0), (0,0,1,0)$$

□

2.C Dimension

THESE PROBLEMS GOT A PUNCH TO EM!!! Well, mainly I struggled on 8. The rest were fine.

1. Show that the subspaces of $\mathbb{R}^2$ are precisely $\{0\}$, all lines in $\mathbb{R}^2$ containing the origin, and $\mathbb{R}^2$.

   Proof. We can split this up by the number of dimensions that the basis of the subspace contains.

   (a) 0 dimensions. The only 0 dimensional vector is $\{0\}$.
   (b) 1 dimension. This is a single vector, and it will always produce a line after scaling in $\mathbb{R}^2$.
   (c) 2 dimensions. Because the 2 dimensions are linearly independent, we know that it spans all of $\mathbb{R}^2$.

   □

3 (a) Let $U = \{p \in \mathcal{P}_4(\mathbb{F}) : p(6) = 0\}$. Find a basis of U.

Proof. If $p(6) = 0$, then all polynomials in U will have a factor of $(x-6)$ (including 0 by technicality). Thus, we can choose the basis

$$\begin{aligned} (x-6) \\ (x-6)(x) \\ (x-6)(x^2)(x-6)(x^3) \end{aligned}$$

□

(b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbb{F})$.

Proof. We can simply add the constant, 1. This is now a linearly independent list which has length $\dim \mathcal{P}_4(\mathbb{F})$, so it is a basis. □

(c) Find a subspace W of $\mathcal{P}_4(\mathbb{F})$ such that $\mathcal{P}_4(\mathbb{F}) = U \oplus W$.

Proof. It is just the subspace given by the span of the additional vector, in this case

$$W = \{a : a \in \mathbb{F}\}$$

□

5 (a) Let $U = \{p \in \mathcal{P}_4(\mathbb{F}) : p(2) = p(5)\}$. Find a basis of U.

Proof. We have a fixed point, which should reduce the dimension by 1 but not in an entirely clear way...

We know there must not be a polynomial of degree exactly 1 because that would make it impossible to satisfy the constraints, so we can simply choose polynomials that satisfy the property. Such as...

$$\begin{aligned} 1 \\ (x-2)(x-5) \\ (x-2)(x-5)x \\ (x-2)(x-5)x^2 \end{aligned}$$

These 3 vectors are clearly linearly independent and are in U. We know that it spans U because it has less than 4 dimensions (otherwise it would equal to $\mathcal{P}_4(\mathbb{F})$) and since this is a linearly independent list of 3 it can't be any lower. □

(b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbb{F})$.

Proof. We can plop in x now. □

(c) Find a subspace W of $\mathcal{P}_4(\mathbb{F})$ such that $\mathcal{P}_4(\mathbb{F}) = U \oplus W$.

Proof. That would be $W = \{ax : a \in \mathbb{F}\}$. □

7 (a) Let $U = \{p \in \mathcal{P}_4(\mathbb{R}) : \int_{-1}^{1} p = 0\}$. Find a basis of U.

Proof. We can create a list of 4 linearly independent vectors. 3 is the maximum basis size for a (strict) subspace of $\mathcal{P}_4(\mathbb{F})$ so we are sure it spans the whole subspace.

$$x$$
$$x^3$$
$$x^2 - \frac{1}{3}$$
$$x^4 - \frac{1}{5}$$

□

(b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbb{R})$.

Proof. We can add just one more linearly independent element, say, 1. □

(c) Find a subspace W of $\mathcal{P}_4(\mathbb{R})$ such that $\mathcal{P}_4(\mathbb{R}) = U \oplus W$.

Proof.

$$W = \{a : a \in \mathbb{F}\}$$

□

8 Suppose $v_1, \ldots, v_m$ is linearly independent in V and $w \in V$. Prove that

$$\dim \operatorname{span}(v_1 + w, \ldots, v_m + w) \geq m - 1$$

Proof. We can subtract the first vector from the rest to obtain that

$$v_2 - v_1, \ldots, v_m - v_1 \in \operatorname{span}(v_1 + w, \ldots, v_m + w)$$

Therefore,

$$\operatorname{span}(v_2 - v_1, \ldots, v_m - v_1) \leq \operatorname{span}(v_1 + w, \ldots, v_m + w)$$

Because

$$\operatorname{span}(v_1, \ldots, v_m) = \operatorname{span}(v_1, v_2 - v_1, \ldots, v_m - v_1) = m$$

We have that

$$\operatorname{span}(v_1 + w, \ldots, v_m + w) \geq \operatorname{span}(v_2 - v_1, \ldots, v_m - v_1) = m - 1$$

□

9 Suppose m is a positive integer and $p_0, p_1, \ldots, p_m \in \mathcal{P}(\mathbb{F})$ are such that each p_k has degree k. Prove that

$$p_0, p_1, \ldots, p_m$$

is a basis of $\mathcal{P}_m(\mathbb{F})$.

Proof. We prove it by induction. First, $\mathcal{P}_0(\mathbb{F})$ is just $\mathbb{F}$. Any non-zero value in $\mathbb{F}$ is a basis for $\mathbb{F}$.

Suppose that $p_0, \ldots, p_i$ was a basis for $\mathcal{P}_i(\mathbb{F})$. Then, we introduce arbitrary $i+1$th degree polynomial p_{i+1}. Given some arbitrary polynomial in $\mathcal{P}_{i+1}(\mathbb{F})$, $f(x)$, we can set the coefficient of p_{i+1} to be the coefficient of the x^{i+1} term of f. Then, what remains after subtracting the p_{i+1} term out is a polynomial in $\mathcal{P}_i(\mathbb{F})$. Because we know that $p_0, \ldots, p_i$ is a basis of $\mathcal{F}_i(\mathbb{F})$, we know we can produce the rest of the polynomial through a linear combination of $p_0, \ldots, p_i$. Thus, $p_0, \ldots, p_{i+1}$ spans $\mathcal{P}_{i+1}(\mathbb{F})$. Because $\mathcal{P}_{i+1}(\mathbb{F})$ has dimension $i+2$ and we have a list of $i+2$ vectors that span it, we know that this list is also linearly independent, and thus we know that $p_0, \ldots, p_{i+1}$ is a basis for $\mathcal{P}_{i+1}(\mathbb{F})$.

Therefore by induction for any $m \in \mathbb{N}$, $p_0, \ldots, p_m$ is a basis for $\mathcal{P}_m(\mathbb{F})$. □

10 Suppose m is a positive integer. For $0 \leq k \leq m$, let

$$p_k(x) = x^k(1-x)^{m-k}.$$

Show that $p_0, \ldots, p_m$ is a basis of $\mathcal{P}_m(\mathbb{F})$.

Proof. We can prove this by observing that these polynomials are actually quite similiar to the ones from last exercise. For example, p_m is a polynomial that contains non-zero terms for degree m only. p_{m-1} has non-zero terms for degree terms for degrees m and $m-1$. Thus, we can use the same argument as the previous exercise, using induction, but this time in reverse. □

11 Suppose U and W are both four-dimensional subspaces of $\mathbb{C}^6$. Prove that there exist two vectors in $U \cap W$ such that neither of these vectors is a scalar multiple of the other.

Proof. The dimension of $U + W$ is at most the dimension of the space itself (6). Therefore, we have that

$$6 \geq \dim(U+W) = \dim(U) + \dim(W) - \dim(U \cap W)$$

$$6 \geq 4 + 4 - \dim(U \cap W)$$

$$2 \leq \dim(U \cap W)$$

Now, we can use 2 basis vectors for $\dim(U \cap W)$ as our pair of linearly independent vectors. □

15 Suppose V is finite-dimensional and V_1, V_2, V_3 are subspaces of V with $\dim V_1 + \dim V_2 + \dim V_3 > 2\dim V$. Prove that $V_1 \cap V_2 \cap V_3 \neq \{0\}$.

Proof.

$$\begin{aligned}
\dim V_1 + \dim V_2 + \dim V_3 &> 2\dim V \\
&\geq \dim V + \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2) \\
\dim V_3 &> \dim V - \dim(V_1 \cap V_2) \\
&\geq \dim(V_1 \cap V_2) + \dim(V_3) - \dim(V_1 \cap V_2 \cap V_3) - \dim(V_1 \cap V_2) \\
\dim(V_1 \cap V_2 \cap V_3) &> 0
\end{aligned}$$

□

17 Suppose that $V_1, \ldots, V_m$ are finite-dimensional subspaces of V. Prove that $V_1 + \cdots + V_m$ is finite-dimensional and

$$\dim(V_1 + \cdots + V_m) \le \dim V_1 + \cdots + \dim V_m.$$

Proof. We prove by induction on m. For $m = 1$, the statement is trivial.

Assume the result holds for $m-1$ subspaces. By the inductive hypothesis, $V_1 + \cdots + V_{m-1}$ is finite-dimensional. Then $V_1 + \cdots + V_m = (V_1 + \cdots + V_{m-1}) + V_m$ is a sum of two finite-dimensional subspaces. By the dimension formula for two subspaces,

$$\dim((V_1 + \cdots + V_{m-1}) + V_m) \le \dim(V_1 + \cdots + V_{m-1}) + \dim V_m.$$

By the inductive hypothesis, $\dim(V_1 + \cdots + V_{m-1}) \le \dim V_1 + \cdots + \dim V_{m-1}$. Therefore,

$$\dim(V_1 + \cdots + V_m) \le \dim V_1 + \cdots + \dim V_{m-1} + \dim V_m.$$

Thus by induction, the result holds for all positive integers m. □

19 Explain why you might guess, motivated by analogy with the formula for the number of elements in the union of three finite sets, that if V_1, V_2, V_3 are subspaces of a finite-dimensional vector space, then

$$\begin{aligned}\dim(V_1 + V_2 + V_3) = &\\ &\dim V_1 + \dim V_2 + \dim V_3 \\ &- \dim(V_1 \cap V_2) - \dim(V_1 \cap V_3) - \dim(V_2 \cap V_3) \\ &+ \dim(V_1 \cap V_2 \cap V_3).\end{aligned}$$

Then either prove the formula above or give a counterexample.

Proof. This statement is FALSE. Consider 3 different lines in $\mathbb{R}^2$. We would have $2 = 1 + 1 + 1 - 0 - 0 - 0 + 0 = 3$! □

20 Prove that if V_1, V_2, and V_3 are subspaces of a finite-dimensional vector space, then

$$\begin{aligned}\dim(V_1 + V_2 + V_3) = &\dim V_1 + \dim V_2 + \dim V_3 \\ &- \frac{\dim(V_1 \cap V_2) + \dim(V_1 \cap V_3) + \dim(V_2 \cap V_3)}{3} \\ &+ \frac{\dim((V_1 + V_2) \cap V_3) + \dim((V_1 + V_3) \cap V_2) + \dim((V_2 + V_3) \cap V_1)}{3}\end{aligned}$$

Proof. This is really 3 formulas squished into one.

We have

$$\begin{aligned}\dim((V_1 + V_2) + V_3) &= \dim(V_1 + V2) + \dim(V_3) - \dim((V_1 + V_2) \cap V_3) \\ &= \dim(V_1) + \dim(V_2) + \dim(V_3) - \dim(V_1 \cap V_2) - \dim((V_1 + V_2) \cap V_3)\end{aligned}$$

We can apply this again for other permutations to obtain three formulas. Summing them together and dividing by three will give the desired formula. I would write it up but I *really* gotta get to chapter 3 now. □

3 Linear Maps

A linear map from V to W is a function $T : V \to W$ with the following properties.

- **additivity** $T(u + v) = Tu + Tv$ for all $u, v \in V$.
- **homogeneity** $T(\lambda v) = \lambda(Tv)$ for all $\lambda \in \mathbf{F}$ and all $v \in V$.

This chapter is all about em!

3.A Vector Space of Linear Maps

Very light on the notes side, probably because this is such a critical new topic. Problems 11 and 17 were basically impossible, I attempted them but it was not possible for me to solve them unfortunately.

1. Suppose $b, c \in \mathbb{R}$. Define $T\colon \mathbb{R}^3 \to \mathbb{R}^2$ by $T(x, y, z) = (2x - 4y + 3z + b, 6x + cxyz)$. Show that T is linear if and only if $b = c = 0$.

   Proof. Suppose that $b \neq 0$. Then, $T(0) = (b, 0) \neq 0$, so its not a linear map.

   Suppose that $c \neq 0$. Then, look at $T((1, 1, 1) + (1, 1, 1)) = T((2, 2, 2)) = (2, 12 + 8c) \neq T(1, 1, 1) + T(1, 1, 1) = (2, 12 + c)$.

   Now let's show that when $b = 0, c = 0$, T is a linear map. Suppose we have $a, b, \in \mathbb{R}^3$.

$$\begin{aligned} T(a+b) &= (2a_1 + 2b_1 - 4a_2 - 4b_2 + 3a_3 + 3b_3, 6a_1 + 6b_1) \\ &= (2a_1 - 4a_2 + 3a_3, 6a_1) + (2b_1 - 4b_2 + 3b_3, 6b_1) \\ &= T(a) + T(b) \end{aligned}$$

   □

2. Suppose $b, c \in \mathbb{R}$. Define $T\colon \mathcal{P}(\mathbb{R}) \to \mathbb{R}^2$ by

$$Tp = \left(3p(4) + 5p'(6) + bp(1)p(2), \int_{-1}^{2} x^3 p(x)\, dx + c \sin p(0)\right)$$

   Show that T is linear if and only if $b = c = 0$.

   Proof. This thing is so bad honestly its easier to prove that the sum of a linear map and a non-linear map is non-linear.

   Suppose we have functions $f \in \mathcal{L}(U, W)$ and $g : U \to W$ (in this case, U and W are just sets who happen to be interpretable as vector spaces). If $f + g$ was a linear map, so would $f + g - f$, which clearly isn't.

   Then, we know that linear maps are preserved under addition, scalar multiplication, and function composition, so everything except the terms with b and c are cleared. $bp(1)p(2)$ is not a linear map (when $b \neq 0$) because it does not preserve scalar multiplication. Multiplying p by a constant multiplites $bp(1)p(2)$ by its square. $c \sin p(0)$ clearly violates both addition and multiplication (sin isn't linear!). So, $b = c = 0$. Since everything else is a linear map, T must be too. □

3. Suppose that $T \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$. Show that there exist scalars $A_{j,k} \in \mathbb{F}$ for $j = 1, \ldots, m$ and $k = 1, \ldots, n$ such that for every $(x_1, \ldots, x_n) \in \mathbb{F}^n$,

$$T(x_1, \ldots, x_n) = \big(\sum_{k=1}^{n} A_{1,k}x_k, \ldots, \sum_{k=1}^{n} A_{m,k}x_k\big).$$

Proof. Consider the standard basis for $\mathbb{F}^n$. The Linear Map Lemma states that a linear map is entirely determined by where it sends its basis vectors. Let

$$Te_i = (A_{1,i}, \ldots, A_{m,i})$$

Then,

$$\begin{aligned} T(x_1, \ldots, x_n) &= Tx_1 + \cdots + Tx_n \\ &= (A_{1,1}, \ldots, A_{m,1}) + \cdots + (A_{1,n}, \ldots A_{m,n}) \\ &= \big(\sum_{k=1}^{n} A_{1,k}x_k, \ldots, \sum_{k=1}^{n} A_{m,k}x_k\big) \end{aligned}$$

$\square$

5 Prove that $\mathcal{L}(V, W)$ is a vector space, as was asserted in 3.6.

Proof. We are given the following definitions for addition and scalar multiplication:

Given $S, T, \in \mathcal{L}(V, W)$ and $\lambda \in \mathbb{F}$,

$$\begin{aligned} (S + T)(v) &= Sv + Tv \\ (\lambda T)(v) &= \lambda(Tv) \end{aligned}$$

Just like the proofs for vector spaces themselves, we can strap our properties to the existing properties of our ring.

(a) Communativity

$$\begin{aligned} (S + T)(v) &= Sv + Tv \\ &= Tv + Sv \text{ (Communativity of } W) \\ &= (T + S)(v) \end{aligned}$$

(b) Associativity

$$\begin{aligned} ((S + T) + H)(v) &= (S + T)(v) + H(v) \\ &= Sv + Tv + Hv \\ &= Sv + (Tv + Hv) \\ &= (S + (T + H))(v) \end{aligned}$$

(c) Additive Inverse
Define $(-T)(v) = -Tv$.

$$(T + (-T))(v) = Tv - Tv = 0$$

(d) Multiplicative Identity
Our identity element is 1. This is trivially the identity, shush.

(e) Distributive property

$$\begin{aligned}(\lambda(S+T))(v) &= \lambda(S+T)(v) \\ &= \lambda(Sv+Tv) \\ &= \lambda Sv + \lambda Tv & = (\lambda S + \lambda T)(v)\end{aligned}$$

□

8 Give an example of a function $\varphi\colon \mathbb{R}^2 \to \mathbb{R}$ such that

$$\varphi(av) = a\varphi(v)$$

for all $a \in \mathbb{R}$ and all $v \in \mathbb{R}^2$, but φ is not linear.

Proof. Maybe something like

$$f(a,b) = \begin{cases} a & a = b \\ 0 & a \neq b \end{cases}$$

would work. □

9 Give an example of a function $\varphi\colon \mathbb{C} \to \mathbb{C}$ such that

$$\varphi(w+z) = \varphi(w) + \varphi(z)$$

for all $w, z \in \mathbb{C}$ but φ is not linear. (Here $\mathbb{C}$ is thought of as a complex vector space.)

Proof. Something like $\varphi(z) = \bar{z}$ would work. It satisfies additivity but

$$i\overline{1+i} \neq \overline{i(1+i)}$$

□

10 Prove or give a counterexample: If $q \in \mathcal{P}(\mathbb{R})$ and $T\colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ is defined by $Tp = q \circ p$, then T is a linear map.

Proof. Not true in general, since something like x^2 violates the additivity constraint. $((1+1)^2 \neq 2^2)$ □

11 Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T is a scalar multiple of the identity if and only if $ST = TS$ for every $S \in \mathcal{L}(V)$.

Proof. I cannot seem to find a good solution to this exercise online, so it is skipped. Maybe I will come back to it. $\square$

12 Suppose U is a subspace of V with $U \neq V$. Suppose $S \in \mathcal{L}(U, W)$ and $S \neq 0$ (which means that $Su \neq 0$ for some $u \in U$). Define $T\colon V \to W$ by

$$Tv = \begin{cases} Sv & \text{if } v \in U, \\ 0 & \text{if } v \in V \text{ and } v \notin U. \end{cases}$$

Prove that T is not a linear map on V.

Proof. Suppose there was some $u \in U$ where $Su \neq 0$. Then, we have some $v \in V, v \notin U$. $u + v \notin U$, so $T(u + v) = 0$. But also $T(u + v) = Tu + Tv = Tu \neq 0$. Contradiction! $\square$

15 Suppose $v_1, \ldots, v_m$ is a linearly dependent list of vectors in V. Suppose also that $W \neq \{0\}$. Prove that there exist $w_1, \ldots, w_m \in W$ such that no $T \in \mathcal{L}(V, W)$ satisfies $Tv_k = w_k$ for each $k = 1, \ldots, m$.

Proof. We have some linear combination, a_i not all 0, such that

$$a_1 v_1 + \cdots + a_m v_m = 0$$

Let a_i be the first non-zero index. We let w_i be an arbitrary non-zero vector in W and all other $j \neq i, w_j = 0$.

Then, we have that

$$T(a_1 v_1 + \cdots + a_m v_m) = 0$$

But also

$$\begin{aligned} T(a_1 v_1 + \cdots + a_m v_m) &= a_1 T(v_1) + \cdots + a_m T(v_m) \\ &= a_1 w_1 + \cdots + a_m w_m \\ &= a_i w_i \\ &\neq 0 \end{aligned}$$

So no such T exists.

$\square$

17 Suppose V is finite-dimensional. Show that the only two-sided ideals of $\mathcal{L}(V)$ are $\{0\}$ and $\mathcal{L}(V)$. A subspace $\mathcal{E}$ of $\mathcal{L}(V)$ is called a two-sided ideal of $\mathcal{L}(V)$ if $TE \in \mathcal{E}$ and $ET \in \mathcal{E}$ for all $E \in \mathcal{E}$ and all $T \in \mathcal{L}(V)$.

Proof. This problem is cooked. $\square$

3.B Null Spaces and Ranges

1 Give an example of a linear map T with $\dim \operatorname{null} T = 3$ and $\dim \operatorname{range} T = 2$.

Proof.

$$\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$$

So V is 5 dimensional. Lets use $\mathbb{F}^5$. Then, we can set

$$T(a_1, \ldots, a_5) = a_1, a_2, 0, 0, 0$$

The range is clearly 2 dimensional, and the null space is clearly 3 dimensional. □

3 Suppose $v_1, \ldots, v_m$ is a list of vectors in V. Define $T \in \mathcal{L}(\mathbb{F}^m, V)$ by

$$T(z_1, \ldots, z_m) = z_1 v_1 + \cdots + z_m v_m.$$

(a) What property of T corresponds to $v_1, \ldots, v_m$ spanning V?

Proof. range $T = V$ corresponds with spanning V. □

(b) What property of T corresponds to the list $v_1, \ldots, v_m$ being linearly independent?

Proof. null $T = \{0\}$ corresponds with linear independence. □

7 Suppose V and W are finite-dimensional with $2 \leq \dim V \leq \dim W$. Show that

$$\{T \in \mathcal{L}(V, W) : T \text{ is not injective}\}$$

is not a subspace of $\mathcal{L}(V, W)$.

Proof. Because T is not injective, we know that $\dim \operatorname{null} T \neq 0$. Thus, given T_1 where $T_1 \neq 0$, let us define some T_2 as

$$T_2(v) = \begin{cases} \text{non-zero value} & T_1(v) = 0 \\ 0 & T_1(v) \neq 0 \end{cases}$$

The specific non-zero value can be defined using basis stuff, but the details are not too important here. What *is* important is that null and range are both subspaces, so by swapping which is 0 and which is non-zero, we will keep these subspaces.

range $T_2 =$ null T_1. T_2 will also be not injective because we swap range and null, and we specify that the range of T_1 must be non-zero. Now the null space of $(T_1 + T_2) = \{0\}$, which is injective, so we have our proof that it is not closed under addition (and therefore not a subspace). □

note Big gap here, on account of the problems in between being too easy.

18 Suppose V and W are finite-dimensional and that U is a subspace of V. Prove that there exists $T \in \mathcal{L}(V, W)$ such that null $T = U$ if and only if

$$\dim U \geq \dim V - \dim W.$$

Proof. Suppose there was a linear map T with null $T = U$. Then,

$$\begin{aligned}
\dim V &= \dim \operatorname{null} T + \dim \operatorname{range} T \\
&= \dim U + \dim \operatorname{range} T \\
&\leq \dim U + \dim W \\
\dim U &\geq \dim V - \dim W
\end{aligned}$$

And the converse is also true because trivial (we can construct using basis stuff) □

19 Suppose W is finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that T is injective if and only if there exists $S \in \mathcal{L}(W, V)$ such that ST is the identity operator on V.

Proof. Suppose that ST was the identity operator. Then, if $v_1, v_2 \in V, v_1 \neq v_2$ and $Tv_1 = Tv_2$, then $STv_1 = STv_2$, which implies $v_1 = v_2$, contradiction.

Suppose that T is injective. Suppose V has basis vectors $v_1, \ldots, v_n$ and W has basis vectors $w_1, \ldots, w_n$ for some subspace of W (we know that $\dim V = \dim W$ because otherwise we would not have an injection). Also, $Tv_i = w_i$ (if it did not map to the basis, it would not be injective, proof is trivial but you use range for it). Then we could also define $Sw_i = v_i$, which is a linear map (trivially), and we are done.

□

21 Suppose V is finite-dimensional, $T \in \mathcal{L}(V, W)$, and U is a subspace of W. Prove that $\{v \in V : Tv \in U\}$ is a subspace of V and

$$\dim\{v \in V : Tv \in U\} = \dim \operatorname{null} T + \dim(U \cap \operatorname{range} T).$$

Proof. First, $\{v \in V : Tv \in U\}$ is a subspace of V. Let's name it X. Let $x_1, x_2 \in X$, then $T(x_1 + x_2) = Tx_1 + Tx_2 \in X$. Same works with scalar multiples.

Next, let's define $T_1 \in \mathcal{L}(X, W)$, $T_1 v = Tv$ for all $v \in X$. By the fundamental theorem of linear maps,

$$\begin{aligned}
\dim X &= \dim \operatorname{null} X + \dim \operatorname{range} X \\
\dim X &= \dim \operatorname{null} T + \dim \operatorname{range} X \\
&\quad \text{(we know this because any } v \text{ that maps to 0 is in } X) \\
\dim\{v \in V : Tv \in U\} &= \dim \operatorname{null} T + \dim(U \cap \operatorname{range} T)
\end{aligned}$$

□

24 Suppose $\dim V = 5$ and $S, T \in \mathcal{L}(V)$ are such that $ST = 0$.

(a) Prove that $\dim \operatorname{range} TS \leq 2$.

Proof. Because $\dim \operatorname{null} ST \leq \dim \operatorname{null} S + \dim \operatorname{null} T$ and $\dim \operatorname{null} ST = 5$, we know that the sum of $\dim \operatorname{null} S + \dim \operatorname{null} T$ is at *least* 5. We also know that $\dim \operatorname{range} TS \leq \min\{\dim \operatorname{range} S, \dim$ ra
The maximum minimum here is clearly 2, so $\dim \operatorname{range} TS \leq 2$. □

(b) Give an example of $S, T \in \mathcal{L}(\mathbb{F}^5)$ with $ST = 0$ and $\dim \operatorname{range} TS = 2$.

Proof. We can shuffle around the basis vector to get a nice non-communative thing going, too lazy to write it out though. $\square$

28 Suppose $D \in \mathcal{L}(\mathcal{P}(\mathbb{R}))$ is such that

$$\deg(Dp) = \deg p - 1$$

for every nonconstant polynomial $p \in \mathcal{P}(\mathbb{R})$. Prove that D is surjective.

Proof. First, we prove that constants get mapped to 0.

Let D_n be D but in $\mathcal{L}(\mathcal{P}_n(\mathbb{R}), \mathcal{P}_{n-1}(\mathbb{R}))$. $D_1 \neq 0$, so it ranges all of $\mathbb{R}$, which is our base case.

Suppose that D_{m-1} was surjective. Suppose $Dm(x^m) = p_m$. Given arbitrary $m-1$th degree polynomial p, since p_m is a $m-1$th degree polynomial, we can find a scalar multiple such that $[x^{m-1}]p = [x^{m-1}]p_m$, lets say its a. Then, we subtract that out of p. What we are left with is a $m-2$th degree polynomial, and since D_{m-1} is surjective and D_m includes D_{m-1}, have some polynomial that maps to $p - ap_m$, and we are done. $\square$

3.C Matrices

The book-proofs in this chapter were a lot harder than previous chapters, but I could still follow them fine.

The problems were *really* easy compared to last subchapter, it makes me wonder what sort of demonic influence that subchapter had...

Highlight to #2, its a great problem!

1 Suppose $T \in \mathcal{L}(V, W)$. Show that, with respect to any choices of bases of V and W, the matrix of T has at least $\dim\big(\operatorname{range} T\big)$ nonzero entries.

Proof. The dimension of the range of T is equivalent to the rank of $\mathcal{M}(T)$, since matrix-vector multiplication is a linear combination of each column of $\mathcal{M}(T)$. Therefore, there is some set of $\dim \operatorname{range} T$ linearly independent vectors. Each of these vectors must not equal zero, so there are at least $\dim \operatorname{range} T$ non-zero entries. $\square$

2 Suppose $T \in \mathcal{L}(V, W)$, where V and W are finite-dimensional and nonzero. Prove that $\dim \operatorname{range} T = 1$ if and only if there exist a basis of V and a basis of W such that with respect to these bases, all entries of $\mathcal{M}(T)$ equal 1.

Proof. If $\dim \operatorname{range} T = 1$, then there is some $w_b \in W$ where every column of $\mathcal{M}(T)$ is a multiple of w_b.

We would like to show there is some basis of W such that $w_b = w_1 + \cdots + w_m$. This itself is equivalent to showing that there is some linear combination that has all non-zero entries (proof will be presented when returning the other direction).

Let us state the lemma we want to prove.

Lemma. *Given vector $v \in V$, $v \neq 0$, there exists a basis $v_1, \ldots, v_n$ such that*

$$v = a_1 v_1 + \cdots + a_n v_n$$

and

$$a_i \neq 0, i \in \{1, \ldots, n\}$$

Proof. Choose $v_1 = v$. We then extend this to a basis, $v_1, \ldots, v_n$. Because for a given linear combination, each element has only one representation, we know that

$$v = v_1 + 0v_2 + \cdots + 0v_n$$

We now define a new basis, $v_1 + \cdots + v_n, v_2, \ldots, v_n$. It remains a basis (unproved but obvious). Using this basis,

$$v = 1(v_1 + \cdots + v_n) - 1v_2 - \cdots - 1v_n$$

None of the coefficients are zero, so we have constructed a solution. □

Now we can proceed with the main proof.

If the entries of the matrix are all 1, it is obvious that $\dim \operatorname{range} T = 1$.

Then, we just need to prove the other way. We are given $T \in \mathcal{L}(V, W)$, $\dim \operatorname{range} T = 1$.

Choose some $v_1 \in V$ where $Tv_1 \neq 0$, and let $Tv_1 = w$. By our lemma, we can construct a basis $t_1, \ldots, t_n$ where $w = a_1 t_1 + \cdots + a_n t_n$, $a_i \neq 0$. We define our new basis, then as

$$w_i = a_i t_i$$

Now,

$$w = w_1 + \cdots + w_n$$

As intended.

As for the rest of the basis of V, we know that

$$\begin{aligned} \dim \operatorname{range} T + \dim \operatorname{null} T &= \dim V \\ 1 + \dim \operatorname{null} T &= n \\ \dim \operatorname{null} T &= n - 1 \end{aligned}$$

So I can define $v_2, \ldots, v_n$ as a basis for null T. We know it is LI from v_1 obviously. So, we can use

$$v_1, v_1 + v_2, \ldots, v_n + v_1$$

as our basis.

For any $j \in \{2, \ldots, n\}$, $T(v_1 + v_j) = T(v_1) + T(v_j) = w_1 + \cdots + w_n$, so our matrix is all ones as desired! □

4 Suppose that $D \in \mathcal{L}\big(\mathcal{P}_3(\mathbb{R}), \mathcal{P}_2(\mathbb{R})\big)$ is the differentiation map defined by $Dp = p'$. Find a basis of $\mathcal{P}_3(\mathbb{R})$ and a basis of $\mathcal{P}_2(\mathbb{R})$ such that the matrix of D with respect to these bases is

$$[D] = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Proof.

$$\begin{array}{c|c} & \begin{matrix} x^3 & x^2 & x & 1 \end{matrix} \\ \begin{matrix} 3x^2 \\ 2x \\ 1 \end{matrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \end{array}$$

I learned blockarray for this! □

12 Prove that matrix multiplication is associative. In other words, suppose A, B, and C are matrices of sizes such that $(AB)C$ is defined. Explain why $A(BC)$ is defined as well, and prove that

$$(AB)C = A(BC).$$

Proof. We can treat A, B, and C as linear maps, in which they are just functions.

Thus,

$$(AB)Cv = A(B(C(v))) = A(BC)v$$

□

13 Suppose A is an n-by-n matrix and $1 \leq j, k \leq n$. Show that the entry in row j, column k, of A^3 (which is defined to mean AAA) is

$$\sum_{p=1}^{n}\sum_{r=1}^{n} A_{j,p}A_{p,r}A_{r,k}.$$

Proof.

$$(AA)_{j,k} = \sum_{r=1}^{n} A_{j,r}A_{r,k}$$
$$(A(AA))_{j,k} = \sum_{p=1}^{n} A_{j,p}(AA)_{p,k}$$
$$(A(AA))_{j,k} = \sum_{p=1}^{n} A_{j,p}\sum_{r=1}^{n} A_{p,r}A_{r,k}$$
$$(AAA)_{j,k} = \sum_{p=1}^{n}\sum_{r=1}^{n} A_{j,p}A_{p,r}A_{r,k}$$

□

16 Suppose A is an m-by-n matrix with $A \neq 0$. Prove that the rank of A is 1 if and only if there exist $(c_1, \ldots, c_m) \in \mathbb{F}^m$ and $(d_1, \ldots, d_n) \in \mathbb{F}^n$ such that $A_{j,k} = c_j d_k$ for every $j = 1, \ldots, m$ and every $k = 1, \ldots, n$.

Proof. When the rank of A is 1, all of the columns are a multiple of a single non-zero column. We can set $c_1, \ldots, c_m$ to be that column, then $d_1, \ldots, d_n$ to be the actual multiples. It is clear this is a biconditional. □

17 Suppose $T \in \mathcal{L}(V)$, and $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$ are bases of V. Prove that the following are equivalent:

(a) T is injective.

(b) The columns of $\mathcal{M}(T)$ are linearly independent in $\mathbb{F}^{n \times 1}$.

(c) The columns of $\mathcal{M}(T)$ span $\mathbb{F}^{n \times 1}$.

(d) The rows of $\mathcal{M}(T)$ span $\mathbb{F}^{1 \times n}$.

(e) The rows of $\mathcal{M}(T)$ are linearly independent in $\mathbb{F}^{1 \times n}$.

Here $\mathcal{M}(T)$ denotes $\mathcal{M}\big(T, (u_1, \ldots, u_n), (v_1, \ldots, v_n)\big)$.

Proof. (a) and (b) are equivalent because any Tv is a linear combination of the columns of $\mathcal{M}(T)$ using the v for the coefficients. Assuming the columns of $\mathcal{M}(T)$ are linearly independent, that makes every resultant vector unique, which satisfies the definition of injectivity. If T was injective, then we can show by contradiction that the columns of $\mathcal{M}(T)$ must be linearly independent.

(b) and (c) are equivalent because $\dim \mathbb{F}^{n \times 1} = n$, so any length n list of linearly independent vectors will also span the whole space, and vice versa.

The same reasoning applies to (d) and (e).

Because row rank and column rank are identical, (c) and (d) are equivalent since the dimensions of both must be n (and therefore they both span the space).

Combining all of these, we have proved the equivalences. □

3.D Invertibility and Isomorphisms

I think it was nice that change of basis was presented before eigenvectors…

The notes for this chapter were pretty long but all the exercises were pretty easy, with none of them that I attempted needing a google or anything.

1 Suppose $T \in \mathcal{L}(V, W)$ is invertible. Show that T^{-1} is invertible and that this inverse is $(T^{-1})^{-1}$.

Proof. Well, we would want

$$T^{-1}(T^{-1})^{-1} = I$$

and

$$(T^{-1})^{-1}T^{-1} = I$$

and T satisfies both, so it is the inverse of T^{-1}. □

3 Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the following are equivalent.

(a) T is invertible

(b) $Tv_1, \ldots Tv_n$ is a basis of V for every basis $v_1, \ldots, v_n$.

(c) $Tv_1, \ldots, Tv_n$ is a basis for V for some basis $v_1, \ldots, v_n$ of V.

Proof. We prove (a) and (b) are equivalent by showing that they are both equivalent to injectivity. (a) is already proven to be equivalent to injectivity.

Suppose that $Tv_1, \ldots, Tv_n$ is a basis for V for some basis $v_1, \ldots, v_n$ of V. Suppose we have $u, v, w \in V$ where $Tv = w$ and $Tu = w$. $T(v-u) = Tv - Tu = 0$. If $v \neq u$, then that implies that we could construct a non-zero linear combination of $Tv_1, \ldots, Tv_n$ that equaled 0, which is a contradiction.

Suppose that T is injective. This implies surjectivity, so each $w \in V$ has some unique $u \in V$ such that $Tu = w$. Given some basis of V, $v_1, \ldots, v_n$, Tv_i forms a basis since there is a unique linear combination for all elements of V, the definition of a basis.

Notice I did not mention any particular basis, which means it works for any and all bases. □

5 Suppose V is finite-dimenstional, U is a subspace of V, and $S \in \mathcal{L}(U, V)$.

Prove that there exists an invertible linear map T from V to itself such that $Tu = Su$ for every $u \in U$ if and only if S is injective.

Proof. Suppose S is not injective. Then T cannot be injective either, and so it also cannot be invertible.

Suppose that S is injective. Let $u_1, \ldots, u_m$ be a basis for U. Then, $Su_1, \ldots Su_m$ is a basis for some subspace of V. Both can be extended and we can construct some basis for V, $u_1, \ldots, u_m, v_1, \ldots, v_n$ where the list $Tu_1, \ldots Tu_m, Tv_1, \ldots, Tv_n$ are bases for V. Then, T will be invertible, so we have constructed a solution. □

6 Suppose that W is finite-dimensional and $S, T \in \mathcal{L}(V, W)$.

Prove that null S = null T if and only if there exists an invertible $E \in \mathcal{L}(W)$ such that $S = ET$.

Proof. Suppose there exists an invertible E such that $S = ET$. For $v \in V$, if T maps v to 0, then so must S because $E(0) = 0$. If S maps v to 0 then so must T because $E(v) = 0$ if and only if $v = 0$ (because it is injective).

Suppose that null S = null T. Consider the basis of null S (or null T, they are the same), $v_1, \ldots, v_n$. We can extend this basis to all of V as $v_1, \ldots, v_n, u_1, \ldots, u_m$. Let the subspace formed from the basis $u_1, \ldots, u_m$ be U.

We will show that S and T are injective in U. Suppose that there is some u_1, u_2 such that $Su_1 = Su_2$. Then, we would have $S(u_1 - u_2) = 0$, but $u_1 - u_2$ is in null S. The only element in both U and null S is 0, therefore $u_1 - u_2 = 0$ and $u_1 = u_2$. Same logic applies to T.

Since T is injective in U, $Tu_1, \ldots, Tu_m$ are linearly independent in W. We can arbitrarily extend this to a basis $Tu_1, \ldots, Tu_m, w_1, \ldots w_r$, where $\dim W = m + r$. We also define a basis $Su_1, \ldots, Su_m, p_1, \ldots, p_r$ using the same principle.

Now we can define E.

$$E(Tu_i) = Su_i$$

and

$$E(w_i) = p_i$$

E is well defined because $Tu_1, \ldots, Tu_m, w_1, \ldots, w_r$ is a basis in W.

E is invertible because $Su_1, \ldots, Su_m, p_1, \ldots, p_r$ is a basis in W. $S = ET$ because the bases all map to the same values (no I will not show it but it is obvious). □

9 Suppose V is finite-dimensional and $T : V \to W$ is a surjective linear map of V onto W. Prove that there is a subspace U of V such that $T_{|U}$ is an isomorphism of U onto W.

Here $T_{|U}$ means that the function T is restricted to U. Thus $T_{|U}$ is the function whose domain is U, with $T_{|U}$ defined by $T_{|U}(u) = Tu$ for every $u \in U$.

Proof. Consider arbitrary basis of W, $w_1, \ldots, w_m$. Because T is surjective, we can find some vectors $v_1, \ldots, v_m$ such that $Tv_i = w_i$ for all $i \in \{1, \ldots, m\}$. $v_1, \ldots, v_m$ is linearly independent, since if it wasn't, $w_1, \ldots, w_m$ wouldn't be a basis. Now we define U as the subspace formed from $v_1, \ldots, v_m$ as a basis. $\dim \operatorname{range} T = \dim W$, and it remains $\dim W$ restricted to U since $w_1, \ldots, w_m$ is basis for W.

$$\begin{aligned}
\dim U &= \dim \operatorname{range} T_{|U} + \dim \operatorname{null} T_{|U} \\
m &= m + \dim \operatorname{null} T_{|U} \\
\dim \operatorname{null} T_{|U} &= 0
\end{aligned}$$

Therefore $T_{|U}$ is an invertible linear map, the definition of isomorphism.

□

12 Suppose V is finite-dimensional and $S, T, U \in \mathcal{L}(V)$ and $STU = I$. Show that T is invertible and that $T^{-1} = US$.

Proof. If T mapped some non-zero vector to 0, there would be no way to recover it back to a non-zero value, so T is invertible. This also shows that U and S are invertible.

The inverse of U is ST and the inverse of S is TU.

$$\begin{aligned}
STU &= I \\
STU(ST) &= ST \\
ST(US)T &= ST \\
S^{-1}T(US)TT^{-1} &= S^{-1}STT^{-1} \\
T(US) &= I \\
US &= T^{-1}
\end{aligned}$$

Therefore we done. □

15 Suppose $T \in (V)$ and $v_1, \ldots, v_m$ is a list in V such that $Tv_1, \ldots, Tv_m$ spans V. Prove that $v_1, \ldots, v_m$ spans V.

Proof. Suppose that $v_1, \ldots, v_m$ was not linearly independent. Then, there is some $a_1v_1 + \cdots + a_mv_m = 0$ with some $a_i \neq 0$. Then, $T(a_1v_1 + \cdots + a_mv_m) = a_1Tv_1 + \ldots a_mTv_m = 0$, which is a contradiction since $Tv_1, \ldots, Tv_m$ is supposed to be a basis. Thus $v_1, \ldots, v_m$ is linearly independent. By the linear map lemma, it also spans V so it is a basis. □

18 Show that V and $\mathcal{L}(\mathbb{F}, V)$ are isomorphic vector spaces.

Proof. Well, $\mathbb{F}$ is just $\mathbb{F}^1$, and any map $T \in \mathcal{L}(\mathbb{F}, V)$ is entirely defined by $T1$, since all other numbers are scalar multiples of $T1$. So we can define our function $S \in \mathcal{L}(\mathcal{L}(\mathbb{F}, V), V)$ as $S(T) = T1$. This function has a clear inverse (that being taking some vector v to the function that maps 1 to v), so we have an isomorphism. $\square$

20 Suppose $q \in \mathcal{P}(\mathbb{R})$. Prove that there exists a polynomial $p \in \mathcal{P}(\mathbb{R})$ such that $q(x) = (x^2 + x)p''(x) + 2xp'(x) + p(3)$ for all $x \in \mathbb{R}$.

Proof. Define $T \in \mathcal{P}(\mathbb{R})$ where $Tp = (x^2 + x)p'' + 2xp' + p(3)$. We want to prove that T is surjective in $\mathcal{P}(\mathbb{R})$.

Consider the basis $1, x, \ldots, x^n$ for vector space $\mathcal{P}_n(\mathbb{R})$. Tx^n is linearly independent from all previous polynomials because it is the only polynomial of degree n. Thus, $T1, Tx, \ldots, Tx^n$ forms a basis, so it is surjective, so we can always construct q. $\square$

24 Suppose A and B are square matrices of the same size and $AB = I$. Prove that $BA = I$.

Proof.

$$\begin{aligned} AB &= I \\ B(AB) &= B \\ (BA)B &= B \\ BA &= I \end{aligned}$$

$\square$

3.E Products and Quotients

These problems all right, yep.

3 Suppose $V_1, \ldots, V_m$ are vector spaces. Prove that $\mathcal{L}(V_1 \times \cdots \times V_m, W)$ and $\mathcal{L}(V_1, W) \times \cdots \times \mathcal{L}(V_m, W)$ are isomorphic vector spaces.

Proof. Consider some map $T \in \mathcal{L}(V_1 \times \cdots \times V_m, W)$. Consider some V_i and have the vector in that position be non-zero and all other vectors in the list be 0. This is clearly equivalent to some $T_i \in \mathcal{L}(V_i, W)$. It also uniquely defines T since $T(v_1, \ldots, v_m) = T(v_1, 0, \ldots, 0) + T(0, v_2, 0, \ldots, 0) + \cdots + T(0, \ldots, v_m)$. Thus we have a invertible map and therefore an isomorphism. $\square$

6 Suppose that v, x are vectors in V and that U, W are subspaces of V such that $v + U = x + W$. Prove that $U = W$.

Proof. If U is a subspace of V and $v \in V$ has $0 \in v + U$, then $v + U = 0 + U$. This is because cosets are disjoint, and $0 + U$ will contain 0, so no other quotient space will contain it.

We have

$$\begin{aligned} v + U &= x + W \\ (v - x) + U &= W \end{aligned}$$

This should be fine since its equivalent to subtracting x from all elements of $v + U$ and $x + W$, which should maintain equality. Since W includes 0, we know that $(v - x) + U = 0 + U$, so $U = W$.

$\square$

8 (a) Suppose $T \in \mathcal{L}(V, W)$ and $c \in W$. Prove that $\{x \in V : Tx = c\}$ is either the empty set or is a translate of null T.

Proof. If $c \notin$ range T, then trivially $\{x \in V : Tx = c\}$ is empty.

If $c \in$ range T, then there is some $v \in V$ such that $Tv = c$. We will show that $\{x \in V : Tx = c\} = v + \text{null } T$. Consider another $w \in V$ such that $Tw = c$. Then, $T(v - w) = Tv - Tw = 0$, so $v - w \in \text{null } T$. Since $w = v - (v - w) \in v + \text{null } T$, we are done. □

(b) Explain why the set of solutions to a system of linear equations such as 3.27 is either the empty set or is a translate of some subspace of $\mathbf{F}^n$.

Proof. Since it is equivalent to the above, obviously. □

9 Prove that a nonempty subset A of V is a translate of some subspace of V if and only if $\lambda v + (1 - \lambda)w \in A$ for all $v, w \in A$ and all $\lambda \in \mathbf{F}$.

Proof. Suppose that A was a translate, $v_1 + U$. Then, for all $v, w \in A$ we can write them as $v = v_1 + u_1$ and $w = v_1 + u_2$. Then, for all $\lambda \in \mathbb{F}$,

$$\begin{aligned}
\lambda v + (1 - \lambda)w &= \lambda(v_1 + u_1) + (1 - \lambda)(v_1 + u_2) \\
&= \lambda v_1 + \lambda u_1 + v_1 - \lambda v_1 + u_2 - \lambda u_2 \\
&= (v_1) + (\lambda u_1 + u_2 - \lambda u_2) \\
&\in v_1 + U
\end{aligned}$$

Suppose that $\lambda v + (1 - \lambda)w \in A$ for all $v, w, \in A$ and all $\lambda \in \mathbb{F}$.

Fix some $w \in A$, and consider any $v \in A$. We will show that $v - w$ forms a subspace of V. For all $\lambda \in \mathbb{F}$,

$$\begin{aligned}
\lambda v + (1 - \lambda)w &= \lambda v + w - \lambda w \\
&= \lambda(v - w) + w
\end{aligned}$$

This looks a lot like the translate formula, if only $\lambda(v - w)$ formed a subspace. This amounts to proving closure under scalar multiplication and vector addition. Scalar multiplication is trivial; λ is right there!

Consider arbitrary $v_1, v_2 \in -v + A$.

$$\begin{aligned}
\lambda(v_1 + v) + (1 - \lambda)(v_2 + v) &= \lambda v_1 + \lambda v + v_2 + v - \lambda v_2 - \lambda v \\
&= \lambda v_1 - \lambda v_2 + v
\end{aligned}$$

Set $\lambda = 1$, and we easily see closure under addition. Thus, A is a translate. □

11 Suppose

$$U = \{(x_1, x_2, \dots) \in \mathbf{F}^\infty : x_k \neq 0 \text{ for only finitely many } k\}.$$

(a) Show that U is a subspace of $\mathbf{F}^\infty$.

Proof. U is closed under addition because if you add 2 vectors with only finitely many non-zero fields, the number of non-zero fields stays finite (since it is at most the sum of the number of non-zero fields of each of the operands). It is also closed under scalar multiplication since scalar multiplication does not change the number non-zero elements (unless it is multiplication by 0, in which is resets it to 0). $\square$

(b) Prove that $\mathbf{F}^\infty/U$ is infinite-dimensional.

Proof. Consider the set of vectors where the ith vector has a 1 in the kp_ith slot for all $k \in \mathbb{N}$, where p_i is the ith prime number. For any pair of vectors in this list, their sum will be multiples of 2 different primes, so they are super linearly independent, and since you would need to add an infinite amount of numbers to get from one to the other, none of these are equivalent in $\mathbb{F}^\infty/U$ either.

This basis is infinite, so $\mathbb{F}^\infty/U$ is infinite dimensional. $\square$

13 Suppose U is a subspace of V such that V/U is finite-dimensional. Prove that V is isomorphic to $U \times (V/U)$.

Proof. Consider each element of V/U, and arbitrarily choose some vector inside the translate to be the "starter" vector, v. Now choose an arbitrary u in this translate. $v-u \in U$, so given some element of $U \times (V/U)$, we can retrieve v and $v-u$. This also entirely determines the vector, and all vectors are covered, and its unique, so its an isomorphism. $\square$

16 Suppose $\varphi \in \mathcal{L}(V, \mathbf{F})$ and $\varphi \neq 0$. Prove that $\dim\big(V/(\text{null } \varphi)\big) = 1$.

Proof. By the rank-nullity theorem,

$$\dim V = \dim \text{null } \varphi + \dim \text{range } \varphi$$

Because we know that $\varphi \neq 0$, we know that $\dim \text{range } \varphi > 0$. But $\mathbb{F}$ only has one dimension, so $\dim \text{range } \varphi \leq 1$. Therefore, $\dim \text{range } \varphi = 1$.

$$\begin{aligned} \dim V &= \dim \text{null } \varphi + 1 \\ \dim \text{null } \varphi &= \dim V - 1 \end{aligned}$$

It is already proven that $\dim V/U = \dim V - \dim U$. So,

$$\begin{aligned} \dim V/\text{null } \varphi &= \dim V - \dim \text{null } \varphi \\ &= \dim V - (\dim V - 1) \\ &= 1 \end{aligned}$$

$\square$

19 Suppose $T \in \mathcal{L}(V, W)$ and U is a subspace of V. Let π denote the quotient map from V onto V/U. Prove that there exists $S \in \mathcal{L}(V/U, W)$ such that $T = S \circ \pi$ if and only if $U \subseteq \text{null } T$.

Proof. Suppose that $U \subseteq \text{null } T$. Then, for all $v_1, v_2 \in v + U$, $Tv_1 = Tv_2$. Then, the choice of S is obvious.

$$S(v + U) = Tv$$

This holds no matter which element we use as the vector in $v + U$ since they're all a difference of some element in null T. Thus we have constructed out map.

Suppose that $U \not\subseteq \text{null } T$. Then, there is some $u \in U \backslash \text{null } T$. Given some vector $v \in V$, we have that $\pi(v) = \pi(v + u)$, but $Tv \neq T(v + u)$. Since S can't differentiate between what was originally v and what was originally $v + u$, S cannot exist. □

3.F Duality

This chapter was way way harder than anything else so far, at least content-wise. Axler even warns of this in the preface, saying that it may be advisable to skip this chapter until it is needed in 9D. I looked at the problems, and the early ones don't look too bad, but I am still skipping them until I cover a later chapter, just so I can ramp back up to the harder ones.

4 Polynomials

Short section on polynomials! The proofs over the fundamental theorem of algebra kinda went over my head, especially the more analysis-y parts, but I guess that's sort of intended. They don't matter for the purposes of these problems anyways.

1 Suppose $w, z \in \mathbb{C}$. Verify the following equalities and inequalities.

For all of these, assume that $z = a + bi$, and that $w = c + di$.

(a) $z + \bar{z} = 2 \operatorname{Re} z$

Proof. $z + \bar{z} = a + bi + a - bi = 2a = 2\operatorname{Re}(z)$. □

(b) $z - \bar{z} = 2(\operatorname{Im} z)i$

Proof.

$$\begin{aligned} z - \bar{z} &= a + bi - a + bi \\ &= 2bi \\ &= 2\operatorname{Im}(z) \end{aligned}$$

□

(c) $z\bar{z} = |z|^2$

Proof.

$$\begin{aligned} z\bar{z} &= (a + bi)(a - bi) \\ &= a^2 + b^2 \\ &= |z|^2 \end{aligned}$$

□

(d) $\overline{w + z} = \bar{w} + \bar{z}$ and $\overline{wz} = \bar{w}\bar{z}$

Proof.

$$\begin{aligned} \overline{z + w} &= \overline{a + bi + c + di} \\ &= \overline{(a + c) + (b + d)i} \\ &= (a + c) - (b + d)i \\ &= a - bi + c - di \\ &= \bar{z} + \bar{w} \end{aligned}$$

□

(e) $\bar{\bar{z}} = z$

Proof.

$$\begin{aligned} \bar{\bar{z}} &= \overline{\overline{a + bi}} \\ &= \overline{a - bi} \\ &= a + bi \\ &= z \end{aligned}$$

□

(f) $|\operatorname{Re} z| \leq |z|$ and $|\operatorname{Im} z| \leq |z|$

Proof. $|z| = \sqrt{(\operatorname{Re} z)^2 + (\operatorname{Im} z)^2}$. Since x^2 is always greater or equal to zero, we have that $|\operatorname{Re} z| \leq |z|$ and $|\operatorname{Im} z| \leq |z|$. □

(g) $|\bar{z}| = |z|$

Proof.

$$\begin{aligned} |z| &= \sqrt{(\operatorname{Re} z)^2 + (\operatorname{Im} z)^2} \\ &= \sqrt{(\operatorname{Re} z)^2 + (-\operatorname{Im} z)^2} \\ &= |\bar{z}| \end{aligned}$$

□

(h) $|wz| = |w||z|$

Proof.

$$\begin{aligned} |wz| &= \sqrt{(\operatorname{Re} wz)^2 + (\operatorname{Im} wz)^2} \\ &= \sqrt{(ac - bd)^2 + (ad + bc)^2} \\ &= \sqrt{(ac)^2 - 2abcd + (bd)^2 + (ad)^2 + 2abcd + (bc)^2} \\ &= \sqrt{(ab)^2 + (ac)^2 + (bc)^2 + (bd)^2} \\ &= \sqrt{(a^2 + b^2)(c^2 + d^2)} \\ &= \sqrt{((\operatorname{Re} w)^2 + (\operatorname{Im} w)^2)((\operatorname{Re} z)^2 + (\operatorname{Im} z)^2)} \\ &= \sqrt{(\operatorname{Re} w)^2 + (\operatorname{Im} w)^2}\sqrt{(\operatorname{Re} z)^2 + (\operatorname{Im} z)^2} \\ &= |w||z| \end{aligned}$$

□

2 Prove that if $w, z \in \mathbb{C}$, then $\big||w| - |z|\big| \leq |w - z|$.

Proof.

$$\begin{aligned} \big||w| - |z|\big|^2 &= (|w| - |z|)^2 \\ &= |w|^2 - 2|w||z| + |z|^2 \\ &= |w|^2 - 2|wz| + |z|^2 \\ &\leq |w|^2 + |z|^2 - 2\operatorname{Re}(wz) \\ &= w\bar{w} + z\bar{z} - w\bar{z} - \bar{w}z \\ &= (w - z)(\bar{w} - \bar{z}) \\ &= |w - z|^2 \end{aligned}$$

Take square roots and we're done! □

4 Suppose m is a positive integer. Is the set

$$\{0\} \cup \{p \in \mathcal{P}(\mathbb{F}) : \deg p = m\}$$

a subspace of $\mathcal{P}(\mathbb{F})$?

Proof. Nope. 2 polynomials can subtract to a non-zero polynomial of smaller degree. □

7 Suppose that m is a nonnegative integer, $z_1, \ldots, z_{m+1}$ are distinct elements of $\mathbb{F}$, and $w_1, \ldots, w_{m+1} \in \mathbb{F}$. Prove that there exists a unique polynomial $p \in \mathcal{P}_m(\mathbb{F})$ such that $p(z_k) = w_k$ for each $k = 1, \ldots, m+1$.

Proof. We can construct a basis for $\mathcal{P}_m(\mathbb{F})$ as follows:

$$p_i(x) = (x - z_1)(x - z_2) \ldots (x - z_{i-1})(x - z_{i+1}) \ldots (x - z_{m+1}) + 1$$

These vectors are linearly independent since they are non-zero at z_i and zero at all other z_j. Since $\dim \mathcal{P}_m(\mathbb{F}) = m + 1$, these vectors form a basis.

We can also construct a polynomial p such that $p(z_k) = w_k$ for each $k = 1, \ldots, m+1$ as follows:

$$p(x) = \sum_{i=1}^{m+1} w_i p_i(x)$$

Since the p_i are linearly independent, this polynomial is unique. □

10 For $p \in \mathcal{P}(\mathbb{R})$, define $Tp \colon \mathbb{R} \to \mathbb{R}$ by

$$(Tp)(x) = \begin{cases} \dfrac{p(x) - p(3)}{x - 3} & \text{if } x \neq 3, \\ p'(3) & \text{if } x = 3 \end{cases}$$

for each $x \in \mathbb{R}$. Show that $Tp \in \mathcal{P}(\mathbb{R})$ for every polynomial $p \in \mathcal{P}(\mathbb{R})$ and also show that $T \colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ is a linear map.

Proof. First we show that $\frac{p(x)-p(3)}{x-3}$ is a polynomial for $x \neq 3$. Suppose that

$$p(x) = a_0 + a_1 x + \cdots + a_n x^n$$

Then,

$$\begin{aligned} \frac{p(x) - p(3)}{x - 3} &= \frac{a_1(x - 3) + \cdots + a_n(x^n - 3^n)}{x - 3} \\ &= a_1 + a_2(x + 3) + a_3(x^2 + 3x + 3^2) + \cdots + a_n(x^{n-1} + 3x^{n-2} + \cdots + 3^{n-1}) \end{aligned}$$

Since each of those terms above has a factor of $x - 3$ (by difference of powers factorization), we can divide out the $x - 3$ (if $x \neq 3$) to retrieve a polynomial. If we show that this polynomial equals $p'(3)$ at $x = 3$ we will be done. Now,

$$p'(3) = a_1 + 3 \cdot 2a_2 + 3^2 \cdot 3a_3 + \cdots + 3^n \cdot na_n$$

When plugging 3 into the equation we get above, we see that they are indeed idenitcal, since all the $x^{n-i-1}3^i$ terms just turn into 3^{n-1}, and they all collect up.

This is a linear map because it simply uses the coefficients of the polynomial and multiplies them by a different polynomial than the standard x^i. It is clearly additive and 0 is mapped to 0.

□

12 Suppose m is a nonnegative integer and $p \in \mathcal{P}_m(\mathbb{C})$ is such that there are distinct real numbers $x_0, x_1, \ldots, x_m$ with $p(x_k) \in \mathbb{R}$ for each $k = 0, 1, \ldots, m$. Prove that all coefficients of p are real.

Proof. We can use the basis from lagrangian interpolation in order to create a polynomial. Since it is a basis, we know it is unique, but langrange polynomials only have real coefficients, so we are already done. □

13 Suppose $p \in \mathcal{P}(\mathbb{F})$ with $p \neq 0$. Let

$$U = \{pq : \; q \in \mathcal{P}(\mathbb{F})\}.$$

(a) Show that $\dim \mathcal{P}(\mathbb{F})/U = \deg p$. (b) Find a basis of $\mathcal{P}(\mathbb{F})/U$.

Proof. Suppose that we have some $g \in \mathcal{P}(\mathbb{F})$, $\deg g \geq \deg p$. Then, by the polynomials division algorithm, we have some $q, r \in \mathcal{P}(\mathbb{F})$ such that

$$g = qp + r$$

Then, $g + U = r + U$. Therefore, all elements of $\mathcal{P}(\mathbb{F})/U$ are equivalent to a translate of degree less than $\deg p$. Additionally, no polynomials with degree less than $\deg p$ can be reduced down further. Therefore, $\dim \mathcal{P}(\mathbb{F})/U = \dim p$.

Such a basis could simply be $1 + U, x + U, \ldots, x^{\deg p - 1} + U$. □

14 Suppose $p, q \in \mathcal{P}(\mathbb{C})$ are nonconstant polynomials with no zeros in common. Let $m = \deg p$ and $n = \deg q$. Use linear algebra as outlined below in (a)–(c) to prove that there exist $r \in \mathcal{P}_{n-1}(\mathbb{C})$ and $s \in \mathcal{P}_{m-1}(\mathbb{C})$ such that $rp + sq = 1$.

(a) Define $T\colon \mathcal{P}_{n-1}(\mathbb{C}) \times \mathcal{P}_{m-1}(\mathbb{C}) \to \mathcal{P}_{m+n-1}(\mathbb{C})$ by

$$T(r, s) = rp + sq.$$

Show that the linear map T is injective.

Proof. Given polynomials r_1, s_1, r_2, s_2 where $T(r_1, s_1) = T(r_2, s_2)$, that either $r_1 = r_2$ and $s_1 = s_2$ or $r_1 \neq r_2$ and $s_1 \neq s_2$. This because if only one were equal and the other disequal, we could subtract and divide everything out into a contradiction.

Suppose there existed some r_1, s_1, r_2, s_2, $r_1 \neq r_2$ and $s_1 \neq s_2$ such that

$$T(r_1, s_1) = T(r_2, s_2)$$

Then

$$r_1 p + s_1 q = r_2 p + s_2 q \tag{1}$$
$$r_1 p - r_2 p = s_2 q - s_1 q \tag{2}$$
$$(r_1 - r_2) p = (s_2 - s_1) q \tag{3}$$

We know that $r_1 - r_2 \neq 0$ and $s_2 - s_1 \neq 0$.

But this means that any zero of p is also a zero of q, and since p is non-constant, there will always be at least one such zero. However, p and q have no zeroes in common, so we have a contradiction! □

(b) Show that the linear map T in (a) is surjective.

Proof. Since $\dim \mathcal{P}_{n-1}(\mathbb{C}) \times \mathcal{P}_{m-1}(\mathbb{C}) = \dim P_{m+n-1}(\mathbb{C}) = m+n$, injectivity implies surjectivity. $\square$

(c) Use (b) to conclude that there exist $r \in \mathcal{P}_{n-1}(\mathbb{C})$ and $s \in \mathcal{P}_{m-1}(\mathbb{C})$ such that $rp + sq = 1$.

Proof. Therefore, since T is surjective, there will always be some polynomials that have $rp+sq = 1$. $\square$

5 Eigenvalues and Eigenvectors

This section narrows our focus even further from linear maps to linear operators, maps from a finite-dimensional vector space to itself.

> To learn about an operator, we might try restricting it to a smaller subspace. Asking for that restriction to be an operator will lead us to the notion of invariant subspaces. Each one-dimensional invariant subspace arises from a vector that the operator maps into a scalar multiple of the vector. This path will lead us to eigenvectors and eigenvalues.

5.A Invariant Subspaces

There are a *lot* of exercises in this subchapter! For this reason I'm solving every 4th problem-ish rather than every problem. Fun fact: this is the first subchapter where I decided not to TEX out problem 1!

3 Suppose $T \in \mathcal{L}(V)$. Prove that the intersection of every collection of subspaces of V invariant under T is invariant under T.

Proof. Suppose we have a collection of subspaces $V_1, \ldots, V_n$ which are all invariant under some $T \in \mathcal{L}(V)$. Now consider some $v \in V_1 \cap \cdots \cap V_n$. What do we know about Tv? Well, we know that $v \in V_i$ for each i because of the invariance of each of these subspaces. Therefore, $Tv \in V_i$ for each i, which means $Tv \in V_1 \cap \cdots \cap V_n$, so we are done. □

4 Prove or give a counterexample: If V is finite-dimensional and U is a subspace of V that is invariant under every operator on V, then $U = \{0\}$ or $U = V$.

Proof. Suppose that U satisfied our condition but did not equal $\{0\}$ or V. Then there is some vector $u \in U$, $u \neq 0$, and $v \in V \backslash U$, $v \neq 0$. Now we can just choose an arbitrary operator where $Tu = v$ and we have a counterexample, and thus a contradiction. □

5 Suppose $T \in \mathcal{L}(\mathbf{R}^2)$ is defined by $T(x, y) = (-3y, x)$. Find the eigenvalues of T.

Proof.

$$\begin{aligned}
T(x, y) &= (-3y, x) \\
\lambda(x, y) &= (-3y, x) \\
\lambda x &= -3y \\
\lambda y &= x \\
\lambda^2 y &= -3y \\
\lambda^2 &= -3
\end{aligned}$$

No real eigenvalues :(. □

13 Suppose $T \in \mathcal{L}(V)$. Suppose $S \in \mathcal{L}(V)$ is invertible.

(a) Prove that T and $S^{-1}TS$ have the same eigenvalues.

Proof. We are given some λ that is an eigenvalue of T. Let's say there is a corresponding eigenvector, $v \in V$. Thus,

$$\begin{aligned} S^{-1}TS(S^{-1}v) &= S^{-1}Tv \\ &= S^{-1}(\lambda v) \\ &= \lambda S^{-1}v \end{aligned}$$

Therefore, $S^{-1}v$ is the eigenvector of $S^{-1}ST$, which has the corresponding eigenvalue which is equal. We can also go in reverse, too.

□

(b) What is the relationship between the eigenvectors of T and the eigenvectors of $S^{-1}TS$?

Proof. They are changes of basis from each other, namely the change S^{-1}. □

16 Suppose $v_1, \ldots, v_n$ is a basis of V and $T \in \mathcal{L}(V)$. Prove that if λ is an eigenvalue of T, then

$$|\lambda| \leq n \max\{|\mathcal{M}(T)_{j,k}| : 1 \leq j, k \leq n\},$$

where $\mathcal{M}(T)_{j,k}$ denotes the entry in row j, column k of the matrix of T with respect to the basis $v_1, \ldots, v_n$.

Proof. Consider the corresponding eigenvector, v, and its representation, $v = a_1v_1 + \cdots + a_nv_n$. Let's consider the basis vector with the largest scalar multiplication, call it v_i with a multiplication of a_i. After applying T, it will have a scalar multiplication of λa_i. But also,

$$\begin{aligned} \lambda|a_i| &= |\sum_{j=1}^{n} \mathcal{M}(T)_{i,j}a_j| \\ \lambda &\leq \sum_{j=1}^{n} \mathcal{M}(T)_{i,j}\frac{|a_j|}{|a_i|} \\ &\leq \sum_{j=1}^{n} |\mathcal{M}(T)_{i,j}| \\ &\leq n \max\{|\mathcal{M}(T)_{j,k}| : 1 \leq j, k \leq n\} \end{aligned}$$

Triangle equality is used here, on the second step. □

23 Suppose V is finite-dimensional and $S, T \in \mathcal{L}(V)$. Prove that ST and TS have the same eigenvalues.

Proof. Consider some eigenvalue, λ for ST, with an eigenvector v. Then

$$\begin{aligned} STv &= \lambda v \\ TSTv &= \lambda Tv \\ TS(Tv) &= \lambda Tv \end{aligned}$$

So λ is an eigenvalue of TS as well, with the eigenvector being Tv.

There's a special case at $\lambda = 0$ but I don't feel like proving it. □

36 Suppose that $\lambda_1, \ldots, \lambda_n$ is a list of distinct positive numbers. Prove that the list $\cos(\lambda_1 x), \ldots, \cos(\lambda_n x)$ is linearly independent in the vector space of real-valued functions on $\mathbf{R}$.

Proof. Consider the vector space given by $\operatorname{span}(\cos(\lambda_1 x), \ldots, \cos(\lambda_n x))$. Consider the double differentiation operator, and its eigenvalues:

$$D^2 \cos(\lambda_i x) = -\lambda_i^2 \cos(\lambda_i x)$$

Therefore, $\cos(\lambda_i x)$ is an eigenvector of D^2, with a corresponding eigenvalue of $-\lambda_i^2$. Since λ_i is always positive, this is guarenteed to be unique. Since each of the vectors is associated with a certain unique eigenvalue, we know they are linearly independent.

□

37 Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Define $\mathcal{A} \in \mathcal{L}(\mathcal{L}(V))$ by

$$\mathcal{A}(S) = TS$$

for each $S \in \mathcal{L}(V)$. Prove that the set of eigenvalues of T equals the set of eigenvalues of $\mathcal{A}$.

Proof. What does it mean for $\mathcal{A}(S)$ to have an eigenvalue? It would be if there existed some $S \in \mathcal{L}(V)$ such that $TS = \lambda S$. In other words, where $(TS - \lambda S) = 0$, so $(T - \lambda I)S = 0$. We know that $S \neq 0$, so there is some $v \in V$ such that $Sv \neq 0$. Applying it, we see that $(T - \lambda I)Sv = 0$, so v is an eigenvector of T with an eigenvalue of λ.

Now we show the other direction. Consider some eigenvalue λ of T such that $(T - \lambda I)v = 0$ for some $v \in V$. Then, we can choose the following function:

$$Su = \begin{cases} u & u = av \text{ for some } a \in \mathbb{R} \\ 0 & \text{otherwise} \end{cases}$$

Since its non-zero, its a valid eigenvector, and its eigenvalue is λ because

$$TSu = \begin{cases} \lambda u & u = av \text{ for some } a \in \mathbb{R} \\ 0 & \text{otherwise} \end{cases}$$

Which means $TS = \lambda S$. □

39 Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T has an eigenvalue if and only if there exists a subspace of V with dimension $\dim V - 1$ that is invariant under T.

Proof. If we have $\dim \operatorname{null} V > 0$, then we have a eigenvalue of 0. Then, we can choose some vector $v_1 \in V$ such that $Tv_1 = 0$. Extend this to a basis, and the subspace generated from $v_2, \ldots, v_n$ is invariant, since its range is still equal to range T. Therefore, from now on we can consider only the situation where $\dim \operatorname{null} V = 0$.

Suppose that V has an eigenvalue, $\lambda \neq 0$, with an associated eigenvector, v_1. Extend v_1 to a basis of V, $v_1, \ldots, v_n$. Now consider the subspace generated from $v_2, \ldots, v_n$, U. Suppose that U was not invariant. Then exists some $v = a_2 v_2 + \cdots + a_n v_n$ such that $Tv = v_1$. But we also know that $T(\frac{1}{\lambda} v_1) = v_1$, and we also know that T is injective from the fact that $\dim \operatorname{null} T = 0$, so we have a contradiction. □

40 Suppose $S, T \in \mathcal{L}(V)$ and S is invertible. Suppose $p \in \mathcal{P}(\mathbb{F})$ is a polynomial. Prove that

$$p(STS^{-1}) = Sp(T)S^{-1}$$

Proof. Firstly, consider that

$$\begin{aligned}(STS^{-1})^n &= \underbrace{(STS^{-1})\cdots(STS^{-1})}_{n \text{ times}} \\ &= S\underbrace{(TS^{-1}S)\cdots(TS^{-1}S)}_{n-1 \text{ times}}TS^{-1} \\ &= ST^nS\end{aligned}$$

Now if we plug each in, we can use this equivalence + distributive property to complete the proof. □

43 Suppose that V is finite-dimensional, $\dim V > 1$, and $T \in \mathcal{L}(V)$. Prove that $\{p(T) : p \in \mathcal{P}(\mathbb{F})\} \neq \mathcal{L}(V)$.

Proof. Polynomials commute, but linear maps of dimension > 1 do not. Therefore, we could not possibly have $\{p(T) : p \in \mathcal{P}(\mathbb{F})\} = \mathcal{L}(V)$. □

5.B The Minimal Polynomial

The proofs in this subchapter were definitely more convoluted than they maybe should have been? But I guess Axler just wants to show off some more "clever" proofs for these…

1 Suppose $T \in \mathcal{L}(V)$. Prove that 9 is an eigenvalue of T^2 if and only if 3 or -3 is an eigenvalue of T.

Proof. Suppose 3 or -3 are eigenvalues of T. Then there is some eigenvector, v, where

$$T(Tv) = T(-?3v) = 9v$$

Next, suppose that 9 was an eigenvalue for T^2. Then there is some v such that,

$$\begin{aligned}(T^2 - 9I)v = 0 \\ (T - (-3)I)(T - 3I)v = 0\end{aligned}$$

So either $(T - 3I)v = 0$, which makes 3 an eigenvalue, or $(T - (-3)I)((T - 3I)v) = 0$, which makes -3 an eigenvalue (with eigenvector $(T - 3I)v$). □

4 Suppose $\mathbb{F} = \mathbb{C}$, $T \in \mathcal{L}(V)$, $p \in \mathcal{P}(\mathbb{C})$, and $\alpha \in \mathbb{C}$. Prove that α is an eigenvalue of $p(T)$ if and only if $\alpha = p(\lambda)$ for some eigenvalue λ of T.

Proof. We have $p(z) = a_0 + a_1 z + \cdots + a_n z^n$.

Suppose we are given an eigenvalue λ, with $\alpha = p(\lambda)$. Consider an eigenvector of T, v where $Tv = \lambda v$. Then $T^k v = \lambda^k v$, so $p(T)v = (a_0 + a_1\lambda + \cdots + a_n\lambda^n) = p(\lambda)v$, which makes $p(\lambda) = \alpha$ an eigenvalue of $p(T)$.

Now suppose that α is an eigenvalue of $p(T)$. Then there is some z where

$$(p(T) - \alpha I)z = 0$$

By the fundamental theorem of algebra,

$$p(T) - \alpha I = c(T - \lambda_1 I)\dots(T - \lambda_n I)$$

Since $p(T) - \alpha I$ is not invertible, at least one of the factors here, $(T - \lambda_i I)$, is not invertible. So there is somt $(T - \lambda_i I)v = 0$, so λ_i is an eigenvalue of T, with $p(\lambda_i) - \alpha I = 0$, so $\alpha = p(\lambda_i)$. □

6 Suppose $T \in \mathcal{L}(\mathbb{F})$ is defined by $T(w, z) = (-z, w)$. Find the minimal polynomial of T.

Proof.

$$\begin{aligned} T^2(w,z) &= (-w,-z) \\ T^2 + 1 &= 0 \end{aligned}$$

□

8 Suppose that $T \in \mathcal{L}(\mathbb{R}^2)$ is the operator of counterclockise rotation by $1°$. Find the minimal polynomial of T.

Proof. T is obviously not a scalar multiple of the identity, so the degree of our minimal polynomial is 2.

$$\mathcal{M}(T) = \begin{pmatrix} \cos(1^\circ) & -\sin(1^\circ) \\ \sin(1^\circ) & \cos(1^\circ) \end{pmatrix}$$

And also

$$\mathcal{M}(T^2) = \begin{pmatrix} \cos(2^\circ) & -\sin(2^\circ) \\ \sin(2^\circ) & \cos(2^\circ) \end{pmatrix}$$

Consider the basis vector $e_1 = (1, 0)$. Then there is some $b, c \in \mathbb{R}$ such that

$$\begin{aligned} T^2 e_1 + aTe_1 + bIe_1 &= 0 \\ (\cos(2^\circ), \sin(2^\circ)) + a(\cos(1^\circ), \sin(1^\circ)) + b(1,0) &= 0 \\ \cos(2^\circ) + a\cos(1^\circ) + b &= 0 \\ \sin(2^\circ) + a\sin(1^\circ) &= 0 \\ a &= -\frac{\sin(2^\circ)}{\sin(1^\circ)} \\ b &= \frac{\cos(1^\circ)\sin(2^\circ)}{\sin(1^\circ)} - \cos(2^\circ) \end{aligned}$$

Wow quite a mess. □

10 Suppose that V is finite-dimensional, $T \in \mathcal{L}(V)$, and $v \in V$. Prove that

$$\operatorname{span}(v, Tv, \ldots, T^m v) = \operatorname{span}(v, Tv, \ldots, T^{\dim V - 1} v)$$

for all integers $m \geq \dim V - 1$.

Proof. Since the left side is a subset of the right side, it is clearly at least as large. Now suppose $m = \dim V$. We would like to show that there is some $a_0, \ldots, a_{m-1}$ such that

$$\begin{aligned} a_0 v + a_1 Tv + \cdots + a_{m-1} T^{m-1} v &= T^m v \\ T^m v - a_{m-1} T^{m-1} - \cdots - a_0 v &= 0 \end{aligned}$$

There is always a solution to this because the degree of the minimal polynomial is less than or equal to m, so we can do a polynomial multiplication to increase the degree to m. □

16 Suppose $a_0, \ldots, a_{n-1} \in \mathbb{F}$. Let T be the operator on $\mathbb{F}^n$ whose matrix (with respect to the standard basis) is

$$\begin{pmatrix} 0 & & & & -a_0 \\ 1 & 0 & \ddots & & -a_1 \\ & 1 & \ddots & & -a_2 \\ & & & & \vdots \\ & & & 0 & -a_{n-2} \\ & & & 1 & -a_{n-1} \end{pmatrix}$$

Here all the entries of the matrix are 0 except all 1's on the line under the diagonal and the entries in the last column (some of which might also be 0). Show that the minimal polynomial of T is the polynomial

$$a_0 + a_1 z + \cdots + a_{n-1} z^{n-1} + z^n$$

Proof. Consider where it throws the first basis vector of $\mathbb{F}^n$, e_0.

$$\begin{aligned} Ie_0 &= e_0 \\ Te_0 &= e_1 \\ &\vdots \\ T^{n-1} e_0 &= e_{n-1} \\ T^n &= (-a_0, -a_1, \ldots, -a_{n-1}) \end{aligned}$$

Since for the ith basis vector, the only 2 items in this list that can touch it are $T^i v$ and $T^n v$, finding each coefficient, given that the coefficient of $T^n v$ is 1, is easy.

$$T^n + a_{n-1} T^{n-1} + \cdots + a_0 T^0 = 0$$

So we are done. □

20 Suppose $T \in \mathcal{L}(\mathbb{F}^4)$ is such that the eigenvalues of T are 3, 5, 8. Prove that $(T-3I)^2(T-5I)^2(T-8I)^2 = 0$.

Proof. This is true if and only if our polynomial is a polynomial multiple of the minimal polynomial of T, so we can prove this instead.

The minimal polynomial has at most degree $\dim V$, so here it is at most 4. Also, it must have a factor of $(T-3I)(T-5I)(T-8I)$ to cover the eigenvalues. It must not contain any other factors that would create a new eigenvalue, so the minimal polynomial is either the above, or the above multiplied by $(T-3I), (T-5I), (T-8I)$. Regardless, the polynomial in our problem remains a multiple of it. □

22 Suppose that V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T is invertible if and only if $I \in \operatorname{span}(T, T^2, \ldots, T^{\dim V})$.

Proof. Suppose that T was invertible. Then the constant term of the minimal polynomial of T (which we will call $p(T)$) is non-zero, call that a_0. Then we can subtract out everything else out of the minimal polynomial to get the identity map.

Suppose that $I \in \operatorname{span}(T, T^2, \ldots, T^{\dim V})$. Then $I = a_1 T + a_2 T^2 + \cdots + a_{\dim V} T^{\dim V}$, so

$$a_{\dim V} T^{\dim V} + \cdots + a_1 T - I = 0$$

Thus this polynomial is a polynomial multiple of the minimal polynomial. Because the constant term is non-zero, the constant term of the minimal polynomial must not be zero as well (because otherwise we would have 0 as a factor of the above polynomial). □

25 Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and U is a subspace of V that is invariant under T.

(a) Prove that the minimal polynomial of T is a polynomial multiple of the minimal polynomial of the quotient operator T/U.

Proof. Let's say that the minimal polynomial of T is p and the minimal polynomial of $T|_U$ is called p_U. Then $p(T|_U)$ clearly equals $0 + U$, so p polynomial multiple of p_U. □

(b) Prove that

$$(\text{minimal polynomial of } T|_U) \times (\text{minimal polynomial of } T/U)$$

is a polynomial multiple of the minimal polynomial of T.

Proof. We will prove that that polynomial sends all vectors to 0 when T is plugged in. Suppose that we have $v \in V$. After applying the minimal polynomial of T/U, all that will be left is some $u \in U$, since $p_{T/U}(v + U) = 0 + U$. After applying the minimal polynomial of $T|_U$, we will be left with nothing. □

29 Show that every operator on a finite-dimensional vector space of dimension at least two has an invariant subspace of dimension two.

Proof. Consider vector space V, an operator $T \in \mathcal{L}(V)$, and its minimal polynomial, $p(T)$. Because we are working in $\mathbb{R}$ or $\mathbb{C}$, $p(T)$ can be factored as

$$p(T) = (T - \lambda_1 I) \dots (T - \lambda_k I)(T^2 + b_1 T + c_1 I) \dots (T^2 + b_m + c_m I)$$

Potentially there are no linear terms, potentially there are no irreducible quadratic terms, but there is definitely at least one of at least one of the 2 types because the polynomial is non-zero.

We will proceed with induction. If $\dim V = 2$, V is trivially our invariant subspace.

Suppose that $\dim V > 2$ and our hypothesis is true for all $n < \dim V$. For some arbitrary vector, when we apply it to $p(T)$, eventually one of them is going send a non-zero vector to the zero vector.

Suppose it was one of the quadratic terms. Then we have some $v \in V$ such that

$$(T^2 + b_i T + c_i I)v = 0$$
$$T(Tv) = -b_i Tv - c_i v$$

Thus, $\operatorname{span}(v, Tv)$ is invariant under T. Its dimension is also 2. If v was a scalar multiple of Tv, then v would be an eigenvector. However, if v was an eigenvector, we would be able to factor out such a factor from the quadratic, contradicting the fact that it is irreducible. Thus, $\operatorname{span}(v, Tv)$ is our desired invariant subspace of dimension 2.

Suppose it was a linear term instead. Then we have some $v \in V$ such that

$$(T - \lambda I)v = 0$$

Thus v is an eigenvector and λ is its eigenvalue. By exercise 39 of Section 5A, this means that we have a subspace, U, of dimension $\dim V - 1$ such that T is invariant under this subspace. Now we can apply our inductive hypothesis to extract out the invariant subspace of dimension 2 for U, which is also an invariant subspace of dimension 2 for V.

□

5.C Upper Triangular Matrices

Yay a shorter subsection! This one is pretty easy, though there are more conditions I need to hold in my head these days.

1 Prove or give a counterexample: If $T \in \mathcal{L}(V)$ and T^2 has an upper-triangular matrix with respect to some basis of V, then T has an upper-triangular matrix with respect to some basis of V.

Proof. Nope. Over $\mathbb{R}$, a 90 degree rotation has no eigenvalues, and thus has no upper triangular matrix. However, 2 90 degree turns forms a 180 degree turn, which does have an eigenvalue of -1, and whose matrix with respect to the standard basis is already upper triangular. □

2 Suppose A and B are upper-triangular matrices of the same size, with $\alpha_1, \dots, \alpha_n$ on the diagonal of A and $\beta_1, \dots, \beta_n$ on the diagonal of B.

(a) Show that $A + B$ is an upper-triangular matrix with $\alpha_1 + \beta_1, \dots, \alpha_n + \beta_n$ on the diagonal.

Proof. Look at it. □

(b) Show that AB is an upper-triangular matrix with $\alpha_1\beta_1, \ldots, \alpha_n\beta_n$ on the diagonal.

Proof. When we see the formula, its all 0s down the row, then $\alpha_i\beta_i$, then all zeroes down the column. □

4 Give an example of an operator whose matrix with respect to some basis contains only 0's on the diagonal, but the operator is invertible.

Proof. The cyclic shift operator is such an operator. Given a basis of V, $v_0, \ldots, v_{n-1}$, the shift operator, T, has $Tv_i = v_{i+1 \pmod n}$ □

5 Give an example of an operator whose matrix with respect to some basis contains only nonzero numbers on the diagonal, but the operator is not invertible.

Proof. The operator with a matrix representation of all 1s is not invertible (it is clearly not injective), but the diagonal is all non-zero (namely, 1). □

7 Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and $v \in V$.

(a) Prove that there exists a unique monic polynomial p_v of smallest degree such that $p_v(T)v = 0$.

Proof. First of all, the minimal polynomial of T, $p(T)$, sends v to zero. So the set of such polynomials is not empty. Therefore, there exists a minimum degree such that such a polynomial exists. Now we prove uniqueness. Consider 2 such polynomials, s, t such that $s(T)v = 0$ and $t(T)v = 0$, both of minimum degree to satisfy that condition. Then $(s - t)(T)v = 0$, and $s - t$ has a smaller degree, so it must be 0. Thus, $s = t$, as desired. □

(b) Prove that the minimal polynomial of T is a polynomial multiple of p_v.

Proof. Consider the minimal polynomial, p. It has degree at least $\deg p_v$. We can use the polynomial division algorithm to split it into q and r such that

$$p = qp_v + r$$

Apply v to both sides,

$$\begin{aligned} p(T)v &= q(T)p_v(T)v + r(T)v \\ 0 &= q(T)0 + r(T)v \\ r(T)v &= 0 \end{aligned}$$

Since r has a smaller degree than p_v and sends v to zero, it must also be zero, so $p = qp_v$ as desired. □

12 Suppose that V is finite-dimensional, $T \in \mathcal{L}(V)$ has an upper-triangular matrix with respect to some basis of V, and U is a subspace of V that is invariant under T.

(a) Prove that $T_{|U}$ has an upper-triangular matrix with respect to some basis of U.

Proof. Consider the minimal polynomial of $T_{|U}$ and the minimal polynomial of T. The minimal polynomial of T is of the form $(T - \lambda_1 I)\dots(T - \lambda_n I)$. The minimal polynomial of T is a multiple of the minimal polynomial of $T_{|U}$, so $T_{|U}$ will be of the form $(T - \alpha_1 I)\dots(T - \alpha_m I)$ where $\{\alpha_1, \dots, \alpha_m\} \subseteq \{\lambda_1, \dots, \lambda_n\}$. Since the minimal polynomial is of that form, $T_{|U}$ has an upper triangular matrix. $\square$

(b) Prove that the quotient operator T/U has an upper-triangular matrix with respect to some basis of V/U.

Proof. The minimal polynomial of T is a multiple of the minimal polynomial of T/U, so the argument is the same as the one above. $\square$